TiLADYS

Public website & Control dashboard

Verified analysis and proposed implementation specification

Review edition 2.0 - 12 October 2026

390	51	52
control contracts	asset briefs	planned acceptance cases

Preserve what already works

Retain the deployed public/control structure, client records, identifiers, current authentication and immutable issued invoice artifacts. Extend the existing system in stages.

Resolve the two design overrides

New invoices: German first, then English. Public site: preserve the actual five-column footer, original wordmark and circuit marks.

Deliver a reviewable specification

This report covers pages, routes, workflows, UI behaviour, database evolution, security, SEO, media and release gates. Detailed machine-readable registers accompany the report. No application code, databases or deployments were changed. Tests are planned, not executed.

The final Codex master prompt and image generation are deferred until the specification is reviewed.

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Version 2.0 - 12 October 2026 - analysis phase only

This is a specification for review, not the Codex master prompt. No application source, database records, settings, credentials or deployments were changed. No new image assets were generated. Read-only repository, hosting and database inspection was performed. This document distinguishes observed facts from proposed behaviour; the included machine-readable registers expand the individual controls, media and acceptance cases.

1. Findings that change the development plan

1. The deployed source is the `feature/services-redesign` branch of `VladyslavTitov/tiladys_portfolio_website`, commit `e82bff04aac9915585b10cc853fe31c558a5964b`. Hosting metadata for both public and control production deployments identifies this commit. `main`, at `9a38f91e9ee92745cc19e90f8beb22ef7f934914`, is substantially older and lacks the CRM/invoice schema. Do not rebuild from main without reconciling the feature branch. [E01, E02]
2. The inspected active default branch of the connected `tiladys-database` project contains Invoice, InvoiceLine, InvoicePayment, Customer, Company, ServiceJob, FileAsset and NumberSequence. The migration history includes completed invoice, recipient and bilingual snapshot migrations. The one-active-draft-per-job unique index exists. The old missing-Invoice-table report is not current for this inspected database. Actual application connection-string identity was not read, so deployment-to-database binding still needs a non-secret environment review. [E03]
3. The PDF renderer explicitly iterates `['en', 'de']`. This conflicts with the latest requirement: **German first, English second**. The code already uses saved bilingual fields, embedded Unicode fonts, page breaking, decimal totals, issuance validation, frozen seller/recipient data and stored issued PDF bytes/checksum. Preserve these foundations. [E04]
4. Contact-photo limits in this deployed source are two files, 1 MiB each and 2 MiB aggregate, with a 4096 dimension and 12 million pixel guard. They conflict with 5 MB each and 15 MB total. Accepting a full 5 MB photo via a Vercel function is incompatible with the documented 4.5 MB function request limit. Direct storage upload with a restricted reservation/finalization protocol is the recommended extension. Merely changing a Next.js body limit does not solve it. [E05, S01]
5. Contact messages and photo attachments are stored together, private; public portfolio reads use an explicit published-only projection. The newer media route rechecks publication on every request, including resizing. Retain these protections. [E06]
6. CRM, service catalogue, customer notes, job line items, invoices and payments exist. Quote, Product, Task, Review, Expense and WebsiteContent tables are absent in the inspected public schema. Their interfaces are proposals, not existing features. No bank reconciliation, external-store sales synchronization, automatic print dispatch or customer portal is verified. [E03]
7. Both hosting projects report Node 24.x and ready production deployments. Metadata reports the execution region `iad1`; the database project is in Frankfurt. The database's reported history window is six hours. Neither region choice nor snapshot history proves a compliant processing arrangement or a complete backup plan. Review processors/transfers and introduce an independently tested recovery policy. [E02, E03]
8. Custom application sessions use random opaque tokens hashed in the database, HttpOnly/SameSite=Strict cookies and production Secure. Credentials are Argon2-based; the static secret phrase is an additional credential, not an independent time-based or possession-based MFA factor. Exact admin-origin validation rejects missing/untrusted origins. The login limiter is an instance-local Map; the newer public contact limiter uses a database counter. Strengthen distributed login abuse protection without replacing functioning authentication. [E07]

9. The live footer is the approved footer. It differs from the mockups: five columns, original wordmark and circuit marks, navy gradient, contact block, social links, legal links and centered copyright. The generated one-row footer is superseded. [E08]
10. The existing SEO layer already has six locale paths, canonicals, hreflang, Open Graph, published-project sitemap entries and slug aliases. Extend it; do not discard it. Existing uk is the Ukrainian language code. Never substitute ua as hreflang. [E09, S02]

Evidence categories and limits

Verified source: exact files at the pinned deployed commit; source inspection is not an execution test. **Verified metadata:** read-only hosting properties, database table/column/index definitions and migration completion records. **Observed live:** public homepage, About/Contact content, footer appearance and public navigation. **Approved concept:** nine public mockup boards, thirteen dashboard boards, the 17-page dashboard kit and tokens. **Proposed:** all behaviour below not explicitly verified above.

No credentials, private customer rows, invoice content or bank details were exported. No application tests, login, mail delivery or production write flow was exercised. Hosted deployment protection can coexist with application login; it must not be disabled for redesign work. Legal guidance endpoints at BMF and Gesetze-im-Internet repeatedly returned errors during verification; exact business-specific tax exceptions and retention periods remain a launch gate. The official federal e-invoicing service confirms the distinction between structured electronic invoices and ordinary image/PDF documents. [S03]

2. Decisions, precedence and scope

Latest explicit user requirements override old generated copy and earlier kit instructions. German-first PDF and the live footer override the old English-first PDF note and every generated footer. Existing factual records override sample names, prices and testimonials. Written requirements and canonical tokens override inconsistent generated labels.

Core release: three-area public navigation/content, preserved portfolio, contact inbox, CRM, jobs, invoice improvements, authenticated Control shell, private uploads, catalogue and publication workflows. **Subsequent core additions:** quotes, product listings/external links and task management. **Optional modules:** receipts/expenses, operational reports, review moderation if real review submissions are introduced, configurable website-copy editing, MFA, automated reminders and monitored backup UI. A disabled optional module must say unavailable/not configured; its button cannot pretend to work.

Single business, owner-managed dashboard is the default. No customer login, public invoice portal, multi-tenant SaaS, marketplace checkout, subscription auto-charge or printer-control connection is authorized by the design. Prepare record boundaries for future roles without building speculative infrastructure.

Items needing confirmation before launch, not before this specification

- Actual business tax treatment, identifiers, approved tax wording, bank/payment details, payment terms and whether structured invoice issuance/receipt is required for each customer type.
- Object-storage provider, budget, regional requirements and permitted model-file formats; proposed first release accepts STL files privately, not G-code/archives.
- Actual available Framer templates, downloadable models, licensing/merchant links and real portfolio/print photos. An empty catalogue is preferable to fake stock.
- Region/processor agreements and retention policy; application-to-Neon binding; recovery credentials/operator and backup retention.

3. Information architecture and routes

Use {locale} = en, de, uk, ru, sk or fr. Preserve currently deployed paths and aliases. Default/x-default remains en unless a separate business decision changes it; the German invoice order is independent of the website default language.

Public page	Proposed canonical route	Data and indexing
Home	/{locale}	Published curated content; index
Service overview	/{locale}/services	Preserve old route; three primary areas; index
Websites	/{locale}/services/websites	Services + website projects + Framer entries; index
3D Printing	/{locale}/services/3d-printing	Services + actual prints + model entries; index
PC Support	/{locale}/services/pc-laptop	Preserve existing path; local/remote services; index
Other digital services	/{locale}/services/business-it	Preserve relevant old services; index
Portfolio	/{locale}/portfolio	Published projects; index; filters are query state
Project detail	/{locale}/portfolio/{slug}	Published only; old aliases redirect directly
Framer catalogue	/{locale}/products/framer	Real published templates; index when substantive
Model catalogue	/{locale}/products/3d-models	Real published models; index when substantive
Product detail	/{locale}/products/{slug}	Real published listing; external purchase default
About / Contact	/{locale}/about, /{locale}/contact	Verified public business information; index
Legal	/{locale}/impressum, /privacy, /terms	Preserve separate real pages; reviewed copy
Legacy prices	/{locale}/prices	Keep or redirect only to an equivalent approved price/service view
Missing/unavailable	original requested URL	Proper 404/410 as appropriate; not soft-200

Primary desktop navigation: Websites, 3D Printing, PC Support, Portfolio, About, Contact, language selector and Start a project. Logo returns to localized Home. Footer retains Home/Services/Portfolio/About/Contact navigation, making the service overview reachable. No indexable route is created for every temporary filter combination.

Dashboard label	Preserved/proposed route	Status
Overview	/dashboard	Existing; extend
Inbox	/dashboard/messages	Existing; extend
Clients	/dashboard/customers, /dashboard/customers/{id}	Existing; UI rename only

Dashboard label	Preserved/proposed route	Status
Companies	/dashboard/companies, /dashboard/companies/{id}	Preserve actual existing routes if different
Jobs	/dashboard/service-jobs, /dashboard/service-jobs/{id}	Existing; extend
Quotes	/dashboard/quotes, /dashboard/quotes/{id}	Proposed
Invoices	/dashboard/invoices, /dashboard/invoices/{id}	Existing API/editor; detail route proposal
Portfolio	/dashboard/projects	Existing; UI rename only
Products	/dashboard/products, /dashboard/products/{id}	Proposed
Tasks	/dashboard/tasks	Proposed list/calendar
Catalogue	/dashboard/prices	Existing; extend structured pricing
Settings	/dashboard/settings/{section}	Proposed grouping of existing billing settings and future settings
Account/Data/Activity	/dashboard/settings/account, /data, /activity	Proposed UI over appropriate existing records
Reviews/Reports	/dashboard/reviews, /dashboard/reports	Optional
Sign-in	/login	Preserve current credentials flow

Private routes, PDFs, files and API payloads are never in public sitemaps. Existing deep links are retained or redirected within the same permission boundary. UI label changes do not require renaming Customer to Client in the database.

4. Canonical visual system

Shared brand and geometry

Navy #071837; cyan #7BE0DF; white #FFFFFF; canvas #F4F7FA; primary text #13243F; secondary text #5C6B82; border #DBE4EE. Success #11734C, warning #8B5100 and error #B42318 are text/icon tones with light tinted backgrounds; states always include words. Proposed font Inter, locally hosted with a system sans fallback. Real vector wordmark/circuit logo remain unchanged.

Spacing scale: 4/8/12/16/24/32/48/64 px. Controls radius 8 px, cards 12 px. Borders 1 px. Standard cards do not need a heavy shadow; menus/drawers may use a restrained navy translucent shadow. Main buttons are cyan with navy text, minimum 44 px height, 16-24 px horizontal padding. Normal controls are 44-48 px high. Focus ring: 2 px dark/cyan high-contrast outline with 2 px offset appropriate to surrounding colour. Essential cyan text on white is avoided because it lacks sufficient contrast.

Public content container proposed max-width 1200 px, desktop side padding 32 px, tablet 24 px, mobile 18 px. Section padding 64-88 px desktop, 40-48 px mobile. Public H1 48-64 px desktop and 32-40 px mobile with 1.05-1.15 line-height; H2 32-40/26-30; body 16-18/16; paragraphs limited to readable line length. Dashboard H1 32, H2 24, body 16, labels 14, captions 12 px. Original dashboard tokens apply to admin typography, not the oversized public hero.

Proposed responsive thresholds: below 768 single-column, 768-1099 adaptive two-column, 1100+ desktop header/sidebar where actual content fits. Do not force a six-link desktop nav at a width that clips it; use the existing menu earlier if needed. Test 320, 360, 390, 768, 1024, 1280 and 1440 px, 200% zoom and long German labels. Existing footer media rules are retained instead of imposing these generic rules on it.

Shared public components

- Header: compact original wordmark, stable top bar; no giant logo. Sticky positioning may follow the existing implementation; reserve its height. Desktop links show the active route. Mobile menu begins immediately below the logo/menu bar, contains a top-aligned list, language choices and contact CTA, scrolls independently, closes on Escape/navigation, restores focus and avoids scroll bleed.
- Hero: navy backdrop, text left and a purpose-specific visual right. Text is HTML, never baked into image. Main visual contains its focal subject in the right half; crop/stack into a 4:3 image below text on narrow screens. No repeated stock laptop on every page. Details/legal pages use shorter heroes; action-heavy Contact starts its form near the fold.
- Service card: illustration or native icon, H3, concise description, one explicit action. A card may have a single stretched link; do not nest clickable buttons/links inside a second button. Three service cards remain equally weighted on Home.
- Portfolio card: authentic screenshot/photo, category, title, short summary, View project. Image ratio 16:10; no private customer name unless publication is authorized. Drafts never enter the public list.
- Product card: preview, product type, price/availability only when factual, View product/Preview/View on Framer. Digital file badge distinguishes model downloads from printed physical objects. External checkout is labelled.
- Gallery: selected image, thumbnail controls with accessible labels, keyboard arrows, zoom/fullscreen Close; image alt describes actual subject. Lazy-load non-leading images. Missing images use a neutral branded placeholder, not a fabricated completed project.
- FAQ: semantic disclosure controls; aria-expanded, keyboard support, readable answers. Every View all questions link has a real FAQ section/route; remove it when no destination exists.
- Final CTA: navy container, real text and two buttons only if distinct. Contact Me opens Contact; Open contact form opens its form anchor. Both may be retained in the live footer-area CTA as approved.

Preserve the live footer exactly in character

Live desktop footer comprises Brand, Contact, Links, Connect and Legal columns. Source CSS uses a 1.25fr/1fr/0.78fr/0.95fr/0.86fr grid, clamp(26px,3.4vw,58px) gaps, 58px top and 22px bottom padding. Its backdrop combines a radial blue accent at 92%/38% with a 110-degree gradient #001025 to #041D3E to #00162F. This is the permitted footer exception to the otherwise restrained flat navy style.

Wordmark display is approximately 215x52 px; brand description max-width 250 px, 14 px with 1.65 line-height; column headings 17 px. Retain subdued circuit marks at approximately 78x58 and 118x88, the contact icons, circular social badges, top/bottom divider and centered copyright. Source responsive rules move to four columns near 1260, two at the existing tablet rule and one centered column below 620; mobile footer padding is 46px 22px 22px with 34px gaps. Preserve this progression, fixing overflow/focus only. Do not reproduce the generated one-row footer.

Verified public contact: contact@tiladys.com, +49 163 7235608, Kronenstraße 19, 45479 Mülheim an der Ruhr. Social destinations: Threads @tiladys.de, WhatsApp +491637235608, Telegram @tiladys_support, Instagram @tiladys.de. Use the deployed business/social source instead of older main-branch email/accounts. Change the brand service subtitle to Websites / 3D Printing / PC Support without changing layout. Legal links remain separate. Circuit marks are decorative with empty alt; linked wordmark alt is TiLADYS. [E08]

Dashboard shell

Desktop sidebar 224 px, navy, compact wordmark and CONTROL label, same order on every page: Overview, Inbox, Clients, Jobs, Quotes & Invoices, Portfolio, Products, Tasks, Settings. Optional Reviews/Reports below More. Highlight current section with cyan/navy high-contrast state. Bottom: View public website, owner initials/avatar and account menu/logout. Real portrait is optional; never use the generated stranger's face.

Header 64-72 px, white, global search with query bounded to accessible records, notifications and interface-language selector. Content padding 24 px desktop and 16 px mobile; canvas #F4F7FA. Cards white, 12 px corners and subtle border. Tables include compact labelled headings, paging and explicit row actions. A 420-560 px adjacent detail panel is useful at large width; below approximately 1100 it becomes a dedicated page/full-screen drawer. No content is hidden by the panel.

Phone layout: top logo/menu bar, list-to-detail Back navigation, stacked cards instead of compressed financial tables, sticky action area with safe-area padding and keyboard clearance. Tablet sidebar may collapse to icons with visible tooltips/labels on focus. Kanban drag is optional and always has an accessible Move to status alternative. Do not use a crowded bottom nav in addition to an equally crowded global sidebar menu; choose one primary navigation treatment.

5. Public page-by-page specification

P01 Home

Order: Header; navy two-column hero; three equal service cards; selected real work; two product pathways; quote process; compact founder teaser; existing footer-area contact CTA; live footer. Hero headline states Websites, 3D prints and PC support, location and online/local availability. Main action opens Contact; secondary View services opens the retained service overview. Visual G01 combines a monitor/site preview, restrained PC chassis and black/white practical prints; it is illustrative, not evidence of client work.

Service cards link to existing website/PC routes and new 3D route. Selected work pulls a bounded published-only collection, category-balanced when available; never invent work to fill three columns. Products link to Framer/model catalogues; hide empty product rows or show an honest availability message. Founder teaser uses real compact portrait R01; More about me opens About. Process is native text/icons. No invented counts, trust badges or response-time claims.

P02 Services overview

Short navy heading, three primary category cards, retained additional digital services and factual catalogue teaser. Customer chooses a topic before navigating; a service-specific Request quote link opens Contact with validated query-state preselection. Preserve /services and legacy /prices intent. Price ranges/from/monthly values must remain clearly labelled and must not be converted directly into agreed invoice amounts.

P03 Websites

Hero with G02 purpose-specific website/device composition; request link selects Websites; Browse Framer templates moves to the template subsection/catalogue. Three service cards Business websites, Landing pages, Updates & maintenance open descriptive detail anchors or preselected enquiry; each link has a distinct defined destination. Gallery uses real portfolio R02-R08. Templates use actual template captures R09, not invented GreenSpace/Fork/Nova products. Four native process steps and focused FAQ follow. View all projects opens /portfolio?category=web-development. Request a website opens Contact. Website gallery empty state offers Contact without claiming experience it cannot show.

P04 3D Printing

Hero G03: visibly layered matte black/white PLA organisers and stand, simple proposition and quote CTA. Three service options: Print from your file, Discuss an idea, Small custom batches. All select 3D enquiry with the relevant request mode. Do not promise industrial precision, all materials or every possible mechanical application. Gallery uses genuine completed print photos R10; model cards use actual owned model previews R11. No generated imagery is labelled completed work.

Material row shows G19 and actual current PLA/colour capabilities; new materials remain enquiry-only until confirmed. Process: send idea/file, confirm feasibility/quote, print and handover. Dimensional/quantity helper is text plus an illustrative sketch/caliper photo G20, without inaccurate raster measurements. User can request with no file. Paid digital files go to configured external merchant; physical printing uses the quote flow. Public model uploads remain disabled until the separate file pipeline is verified.

P05 PC Support

Hero G04 and local/remote mode labels. Describe your issue selects PC topic; Explore services scrolls to service list. Six service tiles: setup/software, troubleshooting, cleaning, SSD/RAM upgrades, backup/data transfer, remote support. Each preselects its service code. Local/remote options open Contact with mode; they do not book an appointment automatically. Device details in enquiry are optional; never ask for account passwords or recovery keys. Remote-help instructions explain that a human agrees a session; no automatic desktop connection. Images G24-G29 are illustrative. FAQ and final CTA use actual availability.

P06 Portfolio and P07 project detail

Portfolio has short hero G05, category controls, optional search, paged real project grid and product-strip links. Category URLs preserve browser Back and can be shared. Keep older Design/Linux/Digital categories under More or All; existing records retain their codes. Add `3d-printing` without migrating all old items into an unrelated bucket. Mongoose, DonetsFit, DonetsFit Calculator, the dual-boot workstation, ETIT/VPU tour, business card and JobAdys remain whenever still published in the live dataset.

Project detail: localized breadcrumb, category/title, actual main image, gallery thumbnails, public overview/role/process/results, technologies, date where factual, optional real website/GitHub links and related projects. Request similar project preselects service and a public project reference. It does not reveal internal job/client IDs. Draft/archived content returns public-unavailable state; authenticated preview has a separate non-indexable private route. Slug aliases redirect in one hop to the current published slug. Templates/models link to their product pages and remain separate entities.

P08 product catalogue and P09 product detail

Use distinct Framer and 3D model category routes. Product cards show real previews, type, availability, price/currency only where verified and View product. Detail: actual visual gallery, description, included files/pages, supported software/print information, licensing summary, and clearly labelled external purchase destination. Preview opens a trusted configured demo URL. If unavailable, remove purchase action and show Contact/Notify only if implemented. External links open with noopener/noreferrer, HTTPS validation, no inferred availability or revenue.

No local customer checkout or payment gateway is in core scope. This avoids confusing dashboard manual invoice records with external product purchases. Optional external sale events require authenticated webhook verification and deduplication; never fabricate synchronization from ordinary URLs.

P10 About

Short hero G06, compact portrait card, founder introduction, principles, factual education, language/communication availability, local/remote cards G21/G22 and contact CTA. Portrait display approximately 240x300 px desktop, max-height 320, restrained crop, no oversized half-page image. Mobile portrait remains small, not full-viewport. Do not publish the generated face. Clearly label in-progress degree; do not imply a completed professional qualification. Website language translations do not prove founder fluency; remove unverified French-speaking claims. Education image G23 is decorative.

P11 Contact and review submission

Reduce the hero height enough to expose the request form early. Hero G07; left form/right contact and 3D helper on desktop, stacked form-first mobile. Fields: topic, request mode where relevant, name, email, optional phone, message, optional photos. Preserve entered body text exactly, including Cyrillic/newlines/emoji; distinguish interface locale from preferred reply language. Inform customers they may write EN/UK/RU; no enforced language detector. German requests are also accepted. Required legal notice wording must match the reviewed processing basis rather than treating marketing consent as necessary for service enquiries.

Photo policy: source files up to 5 MiB each and aggregate 15 MiB, explained as MB consistently in UI, exact byte constants documented. Proposed max 10 files prevents many tiny files; subject to owner confirmation, not the obsolete two-file rule. Validate both per-file and sum. JPEG/PNG/WebP first release; mobile HEIC should be converted in a supported, tested bounded path or receive a specific error, not freeze. Model files have a separate proposed private STL limit (25 MiB each, 50 MiB aggregate, max 3), disabled until storage/resource review. Empty file list is valid. Sketch/photos may be submitted without a model.

Send request displays progress, retains text after errors, gives a reference after persistence and does not say email arrived if only storage succeeded. Rate limits provide Retry-After; server/storage outage gives an actionable error. Footer contact cards use verified mail/tel/social/maps destinations. No unsolicited message or form was submitted during this analysis.

Review form is optional because no existing Review table/route was verified. If introduced, collect display name, actual review text, optional rating and photos; English is a friendly request, not a validation requirement. Publication permission is separate from a private enquiry. Always moderate; photos private until approved. No auto-generated reviews or claims of verified purchase unless true.

P12 Legal, unavailable and search states

Keep Impressum, Privacy and Terms as distinct URLs with readable narrow text (max ~760 px), genuine reviewed copy and the live footer. A giant scales-of-justice hero G08 is optional; recommended default is a compact text-only title. Do not publish generated legal headings or now-obsolete external-dispute links without review. No-results pages state current filters and offer Clear filters; server failures are distinct from genuinely empty results. 404 is a short branded page with Home/Services/Contact, valid 404 status and no fake content.

6. Dashboard page-by-page specification

Canonical page IDs: D01 Overview; D02 Inbox; D03 Clients/Companies; D04 Jobs; D05 Quotes; D06 Invoice list/editor; D07 Invoice detail/payments; D08 Portfolio; D09 Products; D10 Tasks; D11 Catalogue/estimate; D12 Settings/content; D13 Authentication/account; D14 Data/activity; D15 Reviews; D16 Reports. These same IDs are used in the control register and asset bindings.

Every page below inherits authenticated shell, loading/empty/error states, strict server permission checks and no-store caching. Every form field is labelled; client-side validation aids usability but server validation is authoritative. Controls are enumerated in `Control-register.json` with destination, reads/writes, validation, feedback, confirmations and related effects.

D01 Overview

Four cards: new enquiries, active jobs, outstanding issued balance and due tasks. Amounts come from authoritative queries; invoices issued are not money received. Clicking a card opens the filtered section. Job table, new enquiries, Today tasks and follow-up payments provide deep links; quick actions open new-client, quote and job forms. No arbitrary percentage changes from nonexistent historical analytics. When tasks/quotes are not installed, hide their actions instead of generating demo counts.

D02 Inbox

List filters Unread/All/Archived/Spam, bounded search, pagination, original service/language and timestamp. Selecting a message opens detail and explicitly marks read once; a client request cannot set REPLIED. Detail shows original text and authorized photo previews/downloads. UI displays source Contact form; reply locale can be chosen separately. Internal notes are a new private Note relation, not stored over the customer's message.

Create client runs a duplicate suggestion review; Link existing binds only after selection. Create job/quote uses linked client and copies a reviewed brief. Reply by email may retain the existing in-app SMTP action; do not silently switch to mailto or pretend it is a live chat. Composition has explicit Send, SMTP-accepted and failure/unknown states. Replied status follows successful delivery-record handling; preserve existing distinction between delivery failure and sent-but-save-failed. A mail-client link is a separate labelled option. Archive/spam are reversible; no permanent delete as primary action.

D03 Clients and companies

Search/paged table or mobile cards with number, name/company, type, service summary, open jobs, unpaid balance and last contact. Row opens private profile with Overview/Jobs/Quotes/Invoices/Notes/Files/Activity. Add/Edit uses current Customer/Company schema, separate company record and human contact. Default no client is created merely by receiving an enquiry. Confirm duplicates by context; shared family/company email does not force a unique Customer.email constraint.

Billing data can be incomplete for a lead; invoice issuance requires a valid recipient. Profile actions Create job/quote and Add note link to client; View invoice/download file inherits authorization. Archive preserves financial links. Export requires authenticated owner permission, scoped query, auditable action and CSV formula neutralization. Large export is a private expiring download/job, not an overlarge function response.

D04 Jobs and service-specific detail

List/Board switches preserve filters. Do not overload current ServiceJobStatus with quote acceptance or print production. Preserve PLANNED, IN_PROGRESS, WAITING_CUSTOMER, COMPLETED, CANCELLED; add a separate operational stage if Enquiry/Quoted/Printing/Ready views are needed. Quote state belongs to Quote, printing stage to PrintSpec/production record. Board cards derive those stages deliberately; mapping is documented in the workflow register.

Website brief: site purpose, pages, languages, content/assets responsibilities, platform, domain/hosting, milestones, agreed scope. Print details: dimensions with units, quantity, material/colour, model asset, version/hash, orientation notes, estimated hours/grams, feasibility, tolerances agreed with client, inspection and handover. PC details: device/model, stated issue, service mode, accessories received, backup decision and appointment. Never store passwords casually in notes.

Checklist/tasks attach to job; files stay private. Add catalogue lines freezes catalogue label/price mode when agreed, distinguishes internal cost from customer price. Create invoice reuses the unique active draft where present; creating another issued invoice for the same job is a deliberate additional billing decision. Mark ready is a production milestone; Complete is a business completion, neither issues an invoice automatically. Create portfolio draft selects public text/photos, strips confidential detail and requires separate publishing.

D05 Quotes and D06 invoice list/editor

Quotes are new: number, client, optional company/job, line snapshots, validity, exclusions, revision, acceptance evidence and state. Draft may be edited; Sent quote gets a revision record before changed terms are resent. Record acceptance captures date/source/reference; no forged customer signature. Create job/invoice is explicit and idempotent against accepted revision. Expired acceptance requires reviewed renewal.

Invoices reuse current records, catalogue and billing settings. List shows document state (DRAFT/ISSUED/CANCELLED) separately from derived payment state (UNPAID/PARTIAL/PAID) and overdue calculation. Invoice editor fields: recipient type/person/company, billing address, invoice/due/service dates, currency, translated line names/descriptions/units, decimal quantity/rate, tax treatment, service period, notes and payment instructions. Search catalogue supports code/name across relevant translations; selecting a service never silently invoices a 'from' or percentage string.

Save draft validates and retains errors near fields; Preview PDF does not issue or allocate a legal invoice number. Issue invoice shows final review and exact missing requirements, then server locks/validates/freezes/allocates once. All controls and preview UI say German + English, one invoice. Draft company/person toggle clears incompatible company binding and never alters already issued recipient data. Stale browser saves use version/updatedAt conflict detection; the existing row lock already prevents save-versus-issue rewriting.

D07 Invoice detail, payments and correction

Show the frozen legal document, German-first PDF, original total, payment history and live balance. Download issued PDF returns saved bytes, not a regeneration using new client/catalogue/settings values. Partial payments require positive decimal amount <= current balance and locked transaction; existing code has this check. Add payment idempotency and serializable-conflict retry. Credit/overpayment must become a separate reviewed credit workflow, not silently clamped to zero as if no excess existed.

Issued invoices are read-only. Correction/cancellation UI is proposed and must use approved bookkeeping semantics; retain original identifier/PDF/checksum and link correction document, reason, date, actor and affected balance. Cancellation does not delete payment evidence. Reprint/regeneration of legacy issued PDFs is not automatic. A browser-only Confirm modal is insufficient authorization for server issuance.

D08 Portfolio editor

Preserve existing Project/ProjectImage/ProjectSlugAlias and translation data. Category includes new 3D code, old categories available. Cover/gallery use actual media, alt in supported locales, ordered images, published preview. Draft saving and public Publish are separate. The feature branch deliberately removed Project.featured; the design's Featured toggle is not currently backed. Prefer a new explicit curated-placement field/collection if homepage selection is wanted, with additive migration and no restoration of stale old flags without review.

SEO title/description/social image editing reuses existing fields. Rename slug preserves one-hop alias; collision rejects with an explanation. Archive/unpublish immediately hides detail/list/media through authoritative status checks; currently published images must not remain indefinitely publicly cached after withdrawal. Remove image affects only chosen record after confirmation and referential checks. Job-origin images are never made public by editing a caption alone.

D09 Products

Separate Product model, type FRAMER_TEMPLATE/MODEL_3D, Draft/Published/Archived, own slug/aliases/translations, image assets, external preview/purchase URLs, format/license/availability. Exact prices are factual, user-managed and currency explicit. Publishing requires actual product, rights, functioning destination and description. Internal STL originals remain private unless intentionally distributed through an authorized merchant. No arbitrary 'download' endpoint may expose all models. Shared gallery photos can reference media with explicit publication rights; a product is not an issued invoice or finished job.

D10 Tasks and D11 catalogue/estimate

Tasks new: owner, optional customer/job, title/note, due instant and time zone, priority/status, recurrence/reminder only if installed. Calendar and list share the same records; at DST ambiguity ask which time rather than silently shifting. Mark complete can undo within a reversible history; overdue is derived. Browser reminder icon cannot imply a scheduled notification backend.

Catalogue retains PriceSection/PriceItem.code/translations/active/sortOrder. Add structured numeric prices and pricing modes beside existing display strings, never parse a range as an agreed total. Shared searchable picker spans jobs, quotes and invoices. Inactive services remain in historic snapshots; removing an item archives it by default. Bulk save validates each row, reports partial versus atomic policy explicitly; default transaction for moderate edits.

Print estimate separates grams/kg material, machine hours/rate, labour hours/rate, fixed setup and batch quantity. Cost = grams/1000 x material per kg + print hours x machine rate + labour hours x labour rate + fixed costs. Markup price = cost x (1 + markup/100). Gross-margin target instead uses cost/(1-margin/100), with margin <100%. Show per-batch and per-item; add failure allowance/electricity only if not already in machine rate. Transfer estimate to a quote copies reviewed numbers once, does not issue automatically. Example 100g at EUR20/kg, 2h at EUR1/h, 0.5h at EUR20/h, fixed EUR1 gives EUR15 cost and EUR18 with 20% markup before configured tax.

D12 Settings, D13 account/auth and D14 data/activity

Business/invoicing settings preserve actual legal identity, tax mode UNCONFIRMED until owner confirms, identifiers and payment data. Validate display/email/IBAN only as format where applicable; do not invent bank details. Changing settings affects future drafts/issuance, not frozen documents. Number formats currently allocate C/JOB/INV plus business year; reserve advanced custom formats until uniqueness/migration is proven. German invoice primary is a fixed requirement, not a casual reversible locale toggle.

Website content editing is new; preserve file-based copy first or add versioned WebsiteContent JSON incrementally. Editor language tabs en/de/uk/ru/sk/fr, preview and separate publish. Portrait uses real photo, focal crop, desktop/mobile preview. Brand SVG is a trusted owner asset with sanitized import; public upload policy does not accept arbitrary SVG. Optional integrations show Not connected and only enable supported, verified actions.

Account retains current login including secret-phrase field until an intentional migration. Show password inputs, visibility control, generic failures, rate lockout and safe Back-to-login on expiry. MFA uses a maintained TOTP/WebAuthn flow if added, recovery codes handled once; static secret phrase is not labelled MFA. Change credential/session revoke requires authenticated recheck and auditable outcome. Do not add email recovery links without an actual secure recovery endpoint and verified SMTP.

Exports are explicit scoped private downloads with expiry, authorization, status and audit. Backup UI distinguishes not configured, last successful external backup, failed, and restore-test status. Restore opens a review/planned maintenance workflow, not a one-click production overwrite. Audit includes actor/action/entity/time/version/reference but no tokens, full message bodies or bank details. Existing neon_auth tables are present alongside custom AdminUser/Session; do not migrate identity to Neon Auth merely because those tables exist.

D15 Reviews and D16 reports (optional)

Reviews: pending queue, original text/media, explicit publish permission, approve/reject/archive, language tabs only for reviewed translations separate from original, public display-name control. Publishing is a distinct audited transition; quarantine/scanning and photo rights must pass. Reports: issued billed amount, money received, due balance, recorded expenses with periods/currencies distinguished. No fake profit formula from incomparable date bases. Expenses new: actual receipt/date/amount/category/job link; not a tax-deduction certification. No bank-sync or accountant-ready claim until implemented/verified. Negative adjustments require separate records/reasons.

7. Interaction contract

The companion `Control-register.json` is the detailed per-control specification. It resolves shared rules into each row so an implementer can inspect one control without guessing. Every row includes page, label, operation, route/entity, reads/writes, actor, validation, disabled/loading states, success, errors/recovery, confirmation, linked effects, visual/accessibility rules and readiness. Navigation uses links; mutations use buttons; GET never mutates financial data or publishes content.

Shared error taxonomy: 400/422 validation; 401 expired/missing session; 403 authorization/origin; 404 unavailable record; 409 stale/conflicting state; 413 source size; 415 file type; 429 rate limit; 503 dependency unavailable. Specific service codes are mapped to friendly field errors. Preserve data on retry; never display an empty dataset for a server failure. Unexpected errors include a non-sensitive request reference, not a stack trace.

Save/publish/send/issue controls disable only while their operation is pending or prerequisites demonstrably absent, explain why, and re-enable on recoverable error. Repeated clicks share an idempotency key. Search filters debounce 200-300ms, cancel obsolete requests, reset cursor on filter change and preserve Back URL state. Table selection is not a destructive action. Confirm only consequential steps (issue, publish, archive/remove, revoke, restore review, export sensitive bulk data); no modal for every read or draft save.

8. End-to-end relationships and state transitions

Public submissions have no admin credential. A bounded server creates ContactMessage (the enquiry) plus accepted attachment references in one transaction; immutable original text and locale recorded. Storage failure returns failure; acknowledgement mail failure does not undo persistence or trigger duplicate browser submissions. Inbox links a client via the proposed EnquiryLink or a reviewed customerId relation. Matching suggestions do not auto-merge. A job/quote references its source enquiry and customer; private brief may be copied but original remains unchanged.

Quote Draft -> Sent -> Accepted/Declined/Expired. Sent revision changes create a new revision; acceptance records exact revision. Conversion creates at most one linked job for the same conversion key, while additional jobs are explicit. Job current statuses are preserved; production stage and quote milestone are separate. Completion can offer a publication draft but does not publish or invoice automatically.

Invoice DRAFT -> ISSUED -> CANCELLED only through supported reviewed transition. Payment state is computed from active payment ledger; overdue = issued unpaid balance >0 and due local business date passed. A PAID badge is not a document status. Existing row-level locks, atomic NumberSequence and partial draft index are retained. New idempotency ensures retry after a lost response returns the same payment/conversion rather than creates a second one.

Public Project/Product/Review uses Draft/Pending -> Published -> Archived/unpublished with explicit owner action. Public response contains only approved projection and published media. A private FileAsset never acquires visibility merely because its linked job completed. Publish selection copies a sanitized derivative into public media with rights/consent metadata and audit link to source. Backend/cache invalidation checks are part of publication.

9. Database proposal and safe evolution

Preserve current entities

AdminUser/Session/AuditLog; Customer/Company/CustomerNote/CustomerActivity; ServiceJob/ServiceJobLineItem; PriceSection/PriceItem; BusinessBillingSettings; Invoice/InvoiceLine/InvoicePayment; NumberSequence; Project/ProjectImage/ProjectSlugAlias; FileAsset; ContactMessage/ContactAttachment/ContactRateLimit; VisitorDay. Preserve IDs, stored data, published slugs and frozen invoice bytes. Existing numeric money scales are 12,2 or 14,2; quantities 12,3. Public schema also contains managed neon_auth tables in a separate schema; those are not the active application auth source verified in code.

Additive proposed entities/fields

Entity	Fields and relationships	Constraints/indexes
Quote / QuoteRevision / QuoteLine	customer/company/job optional; number; revision; snapshots; validity; accepted evidence	unique number/revision; indexed customer/status/date; no mutable accepted revision
QuoteAcceptance	exact QuoteRevision FK; accepted date; source/reference; recording actor; actual evidence asset optional	one authoritative acceptance per revision; evidence private; correction history append-only
Product / ProductAlias	type; slug; translations; license; price/currency; merchant/demo URL; status; media	unique slug/alias; status/type/order index
Task	optional job/customer; title; note; due instant; zone; owner; priority/state	due/state and job indexes; FK restrict/archive policy
PrintSpec	1:1 job; dimensions numeric+unit; quantity; material/colour; selected model hash; estimates; feasibility; stage	unique jobId; positive bounds/checks; selected file belongs to job
WebsiteBrief / PcBrief	1:1 job structured service fields; private details	unique jobId; safe schema version
EnquiryLink / EnquiryNote	original ContactMessage -> customer/job/quote links; author/time	unique conversion key; indexed source/customer
UploadReservation / MediaObject	actor/source form; limits; random key; size/hash; quarantine/accepted state; expiry; derivative metadata	unique key; expiration/status index; claimed once
PublicationSelection	source private file -> approved derivative/project/product/review; rights record	public whitelist projection; audit actor/date
Review	original text; display name; topic; pending/published state; permission; media	state/date; never publish email/IP
Expense	date; amount/currency; category; receipt; optional job; reason	date/category/job indexes; separate reversal record
WebsiteContent / Revision	locale/key/JSON schema; revision; published revision	unique key/locale/revision; versioned publish
IdempotencyRecord	actor/scope/key/body hash/result identity/status/expiry	unique(actor,scope,key); conflict on key reused with changed payload
EmailOutbox / DeliveryAttempt	source action; dedupe key; recipient reference; queued/accepted/failed/unknown	unique action key; retry/status/time index
PaymentReversal / InvoiceCorrection	links original record; reason; actor; immutable correction artifact	unique reviewed action; retention rules, not hard delete

New mutable business entities use stable opaque IDs, createdAt/updatedAt, an explicit revision for stale-write checks and archivedAt where applicable. Immutable revisions, ledger events and delivery attempts retain createdAt/occurredAt and recording actor/source; use append-only correction rather than

overwriting history. Single-owner access is enforced server-side; future staff ownership/role policy is an explicit extension, not a silent tenant schema. Financial relationships use restrictive deletion and archival; all dates have defined UTC-instant versus Europe/Berlin business-date semantics.

Customer.email remains indexed, not globally unique. A household/company can share email. Company contact joins may be added when a real need appears; do not duplicate billing snapshots under client names. Avoid polymorphic unvalidated IDs; either explicit FKs or a checked binding model. Public entity projections never use unrestricted `include` for related customer data.

Numeric and date rules

API accepts decimal strings with one normalized separator; reject ambiguous grouping or exponential/out-of-range forms. Server computes HALF_UP line rounding consistently with current code and verified tax profile. Persist units, price modes, agreed rate and period. Return decimal strings, not JS float arithmetic. Correct totals from saved line snapshots; UI recalculation is preview only. Calendar dates are business-date values, appointments UTC instants plus zone; preserve historic timestamp interpretation and do not bulk convert without mapping. Numbering uses Europe/Berlin year, atomic counter row and unique issued number; no SELECT MAX+1. Monotonic values may include documented gaps; never reuse an issued number.

Connection, roles and files

Retain Prisma and existing pooled driver; use the provider-recommended direct migration/dump connection and dedicated migration role. Public runtime needs SELECT published content and narrowly defined contact/upload writes, not all CRM privileges. Control runtime gets necessary DML, no DDL/role management. Credentials server-only, separate by environment; never in NEXT_PUBLIC variables. Define grants against actual query paths; public app currently writes contact directly, so a SELECT-only role alone would break it. Use transaction-local context if later RLS is introduced; transaction pooling must not rely on session-wide SET. Application authorization remains necessary.

Existing BYTEA attachments/project images/PDFs can remain while migration is staged. Proposed object storage takes original/large files and derivatives; keep DB metadata/checksum/references. Never store files in an ephemeral deployment filesystem. Private downloads authorize every request and issue short-lived scoped URLs only when safe; object-key secrecy is not authorization. Database and object backup recovery must be consistent.

Migration and rollback sequence

1. Verify deployed commit, runtime, current schema, migration history and non-secret environment mapping. Capture counts/checksums privately; do not include client data in the handoff.
2. Take a recoverable database backup and corresponding object manifests; verify decryption/access and restore onto an isolated test database. Six hours of provider history is not the independent backup policy.
3. Use an isolated schema-only/sanitized test environment for new modules or a controlled production-like branch with access restrictions. Do not expose customer data in public preview branches.
4. Generate reviewed additive Prisma migrations; add nullable/version fields, new relations and indexes first. Preserve named existing partial indexes/constraints absent from ORM representation.
5. Run deterministic backfills in bounded batches with checkpoints, validated translated fields and explicit unknown states. Do not infer tax data/consent or renumber old invoices.
6. Deploy compatibility readers/writers behind module configuration; test and compare counters/links/bytes. Enable the new interface only after data checks.

7. For public media migration, write derivatives, verify hashes and rights, then switch reads; retain old private bytes until verified retention/purge decision.
8. Roll back app version/feature selection on failure; do not roll back the whole live database over new valid writes. Destructive down-migrations and `migrate reset` are excluded from normal production work. Restore requires a reviewed cutover plan.

10. Invoice preservation and German-first extension

Verified existing mechanics

The feature branch stores seller, recipient, bank/payment, tax and bilingual line fields on Invoice/InvoiceLine, documentVersion, issuedPdf bytes/checksum/size. Issuance uses a transaction and Invoice row FOR UPDATE, checks tax/business confirmation and translations, allocates through an atomic scope/year upsert, generates the PDF and records audit/activity. Draft PATCH locks the same row and rejects non-DRAFT to protect against delayed editing. Payment creation locks and rejects amount >balance inside a Serializable transaction. PDF GET requires currentUser and returns stored issued bytes with private/no-store headers. The database has the partial unique index for one current draft per linked job. [E04, E03]

Required change, not a fresh invoice rewrite

New renderer begins de, then en; labels, preview tabs and tests follow that order. Introduce a document/layout version (proposed 3) and explicit language order in new snapshot metadata. German section is the original presentation; English label: English translation of the same invoice. No additional payment is due. Remove the old implication that German is only the second translation. One snapshot feeds both language sections; translations are frozen on issue. Keep equivalent names/address/identifiers/amounts unchanged, including culturally sensitive names. Avoid translating banking identifiers.

Historical issued PDF bytes/checksum/order remain untouched by this rollout. New draft previews and newly issued documents use German-first. If an owner needs a revised presentation of a historical issued document, create a separately auditable rendering linked to the same snapshot/version under the reviewed accounting policy; never silently replace original bytes. A changed locale query may change UI labels, not regenerate an issued legal document.

Translation and pagination

Both DE and EN customer-facing fields must be present before issue; no machine-generated translation is accepted silently. Existing generic `serviceName`/`description` fields are not proof of English because catalogue selection can prefer German; preserve them for compatibility but validate or add explicit EN fields and label the editor. Existing `...De` fields remain. Optional empty notes need no fabricated counterpart. Missing translation blocks issue with a field link; draft preview may show a clear placeholder only in DRAFT.

A short document is two complete language sections on pages 1 and 2. Long German may be pages 1-3, English begins page 4; per-section and global page indicators may be added. Repeat table headings on continuation pages, wrap names/descriptions, paginate a single very long row in bounded chunks, keep totals/payment instructions readable and avoid 6 px microtext. A4 portrait with ~40-48 pt margins, embedded Unicode fonts; verify EUR, umlauts, Cyrillic names and mixed-language long addresses. Raster hero photos are unnecessary in invoices. Use original vector logo or sharp trusted raster derivative.

Validate bank/tax settings through the existing unconfirmed gate. Do not hardcode mockup 19%, 0% or fabricated statements. For a small-business exemption, the correct display wording/rate rows depend on confirmed legal treatment, not a decorative zero-tax dropdown. Address, service period, recipient and due-date requirements are configured and validated against current rules. Current official tax-specific verification was incomplete due source access errors, so legal completeness is not claimed. A normal bilingual PDF is not by itself a structured e-invoice; design a separate exporter/receipt storage capability if applicable. [S03]

11. Security model and verification responsibilities

This is a proposed target plus verified code observations, not a certification. The main boundary is public browser -> limited public API -> validated records, versus owner browser -> authenticated private API -> CRM/financial/files. Storage, mail and external merchants are separate trust boundaries. Attacker goals include spam/storage exhaustion, account takeover, identifier-based file/record exposure, forged mutations, malicious image/model inputs, stale/concurrent financial writes and compromised external links.

Area	Required protection and placement	Evidence/check
Authentication	Preserve Argon2 credential verification and hashed opaque session tokens; generic failures, distributed IP/account throttling, rotation/expiry/revocation. Do not label static phrase MFA.	Source E07; test expired/revoked token and parallel login limits
Authorization	Central per-route owner policy and relation checks for all customer/job/invoice/media IDs; deny by default. Publication reads use explicit projection. Future staff permissions attach to operations, not menu visibility.	Anonymous and unauthorized GET/PATCH/download tests [S04]
CSRF	Exact configured origins for cookie-auth mutations; missing/untrusted origin rejection where browser flow guarantees it; explicit CSRF token if architecture requires exceptions. GET side-effect-free. SameSite complements protection.	Existing trusted-origins; forged-origin and missing-origin tests [S05]
Session/cookies	HttpOnly, Secure production, host-only Path=/ cookie, appropriate SameSite, bounded idle/absolute lifecycle; no localStorage auth secrets. Public/control origins considered separately.	Current 8-hour session observed; logout/revoke/expiry tests [S06]
XSS/content	Escape contact/notes/filenames; allowlisted sanitized rich text only if required. JSON-LD safe serialization, no unsafe inline customer HTML. CSP designed for actual resources; no blanket unsafe-eval fix.	Stored/script payload appears as text; CSP report review
SQL/data	Prisma parameterized queries and safe tagged SQL; never interpolate user identifiers; bounded filters, row/field allowlists, decimal constraints.	Existing tagged SQL; SQL-injection/overposting tests
Uploads	Reservation bound to purpose/draft; 5 MiB/15 MiB server checks; type signature and image decode/re-encode, pixel/time/memory limits, random keys, quarantine, authorized finalization.	Existing sharp guards retained; malformed/oversized/finalization tests [S07]
Model files	Private STL only initially, resource-bounded validation; no automatic machine execution, G-code/archives disabled. Never send confidential designs to public scanners.	Scanner sandbox and file association checks

Area	Required protection and placement	Evidence/check
Downloads/PDFs	Auth every private access, relation checks, no-store, nosniff, safe Content-Disposition, short-lived scoped storage access. Public media checks published state.	Cross-message file-ID tests and draft media denial
SSRF	Prefer uploaded images; remote URL fetch only from narrow HTTPS allowlist with resolved-address checks, no private/link-local targets, controlled redirects, byte/time limits.	Mock/test controlled local targets denied [S08]
Secrets	Server-only per-environment keys/roles; no frontend bypass tokens, broad database credentials or secrets in logs/archive. Build output and source map review.	Secret scan and frontend bundle review
Rate limits	Database/managed shared counters with bounded keys/TTL; trusted source IP extraction and global budget; login limiter Map insufficient across replicas.	Distributed instance test; legitimate shared-IP fallback
Headers/CORS	Control frame denial, nosniff, referrer/permissions policy, appropriate CSP/HSTS after HTTPS/domain review; exact CORS public origin if needed. CORS is not authorization.	Browser/header check, no wildcard credential CORS
Mail/external events	Outbox/idempotency; safe text; no arbitrary SMTP recipients/headers; authenticated webhook signatures and replay window if merchant integration added.	SMTP accept vs delivery vs record failure tests
Private cache/logs	Private DTOs no-store; public cache keyed only by published revision; redact secrets/message bodies/PII. Audit critical mutations by owner.	CDN/public response leakage and logging review
Database/backups	Least-privilege runtime vs migration roles, TLS verification, restricted management access, encrypted independent backups, known restore operator/test.	Provider policy + restore drill; no claimed backup without evidence
Dependencies/build	Pinned lockfile, maintained validation/image/PDF libraries, dependency advisories/CI checks, scoped previews.	Build/tests/advisory review at implementation time

Direct-upload protocol for the larger photo policy

Public browser validates source size/count and creates a bounded metadata request. Server issues an expiring reservation allowing only purpose-specific random keys, exact maximums and short TTL, after abuse controls. Bytes go directly to selected storage instead of through the 4.5 MB Vercel function. Original upload is quarantined/private. A bounded processing worker verifies actual stored byte counts, signature, dimensions, pixel count and decode/re-encode limits; derive safe previews. Finalize consumes reservation once, matches claimed files/hashes/purpose, enforces aggregate original-byte cap and atomically links accepted attachments to one enquiry. A failure leaves the message unsent or accepted without photos only after explicit customer choice; orphan reservations expire and are cleaned up. Malware-scan failure is pending/rejected, never 'safe by default'.

Unauthorized callers cannot attach somebody else's reservation, public object key or private client file. Late finalize and repeated requests return known status. Direct upload is not a security bypass: the app verifies post-upload objects independently. Limit storage/processing cost per IP/session/purpose, with fair recovery for shared networks. Do not expose public list/read privileges on private buckets. [S01, S07]

Privacy, processing and residual risk

Collect only information needed for requests/work, define processing purpose/legal basis, retention and public-publication permission separately. Do not require service enquiry consent to marketing. Explain processors, optional analytics/maps and external purchase links; a static map link avoids third-party embed loading until requested. Preserve original message while displaying it safely. Set legal/business retention policy by record category; financial records are not deleted by a generic GDPR button when retention applies. Private exports/receipts and image EXIF require careful handling. [S09]

Hosting Frankfurt data with US execution metadata means processor/transfer review is needed; geography alone is not a legal conclusion. App authorization failure, compromised owner account, undiscovered parser flaw, external merchant outage and backup gaps remain residual risks. Keep blast radius small, monitor failures and test recovery. Production branch was reported unprotected and IP allowlist empty; this is not proof of compromise. Investigate compatible restrictions without blocking public contact writes or disabling current deployment protection.

12. SEO, multilingual content and performance

Preserve /en,/de,/uk,/ru,/sk,/fr route language and set HTML lang correctly. Use self-canonical URLs per actual language, reciprocal hreflang only for available equivalent published pages, and x-default for chosen default. Keep locale switch on equivalent page where possible; if translation unavailable, say so and navigate deliberately, not silently to wrong content. Avoid bulk indexing English fallback under six language tags. Existing generic UI-copy entries must be translated accurately; customer submitted language is independent. [E09, S02]

Sitemap includes 200 canonical public pages, substantive product catalogues/details and published projects; excludes drafts, aliases/redirects, filters, private previews, APIs and dashboard. lastmod reflects content modification, not 'now on every request'. Keep legacy prices/service/project mapping and one-hop slug aliases; use 301/308 for permanent moves when suitable. Publication invalidates the public list/details/sitemap consistently. If database temporarily fails, preserve static pages and identifiable safe published cached results; don't replace a rich site sitemap with misleading brand-new timestamps. [S10]

Each service page has a distinct H1/title/description, factual local context and specific work, not interchangeable generic IT text. Default titles may combine service + TiLADYS + Mülheim where relevant. Use genuine Organization/LocalBusiness information and Service/BreadcrumbList; Product/Offer only for actual available purchasable items and real prices/merchant relationship. No invented AggregateRating or self-serving star promises. Social image 1200x630 with rendered HTML/vector text and approved visual; don't bake a guessed logo in AI imagery. Published portfolio updates preserve SEO/social fields.

Private routes and PDF/file responses use auth plus no-store; add noindex headers/meta as an additional indexing hint. Robots exclusion does not secure private content and can prevent a crawler seeing noindex; do not make private records public to help indexing removal. Never list sensitive identifiers in public robots URLs. [S11]

Performance targets are proposed acceptance budgets, not measured current results: LCP <=2.5 s, INP <=200 ms, CLS <=0.1 at field 75th percentile when enough data exists; laboratory testing complements, not proves field results. Initial above-fold transfer target <=1 MB compressed, hero ~150-250 KB at rendered size where feasible, card ~40-100 KB, responsive srcset/sizes, width/height reserved. Lazy load below-fold media; preload only the actual LCP visual/font. No autoplay heavy hero videos, multiple 4K downloads or PDF fonts on every dashboard route. Virtualize only when counts justify it; use pagination for CRM. [S12]

Object derivatives can be versioned/cacheable if publication can withdraw access promptly; sensitive/status-controlled media should retain authoritative access checks and private/no-store until a safe cache/purge design is verified. Image processing should not run full database-BYTEA decode on every public resized request indefinitely; precomputed bounded variants are a proposed efficiency improvement. Static footer/header icons are original SVG/Lucide, not network image generation.

13. Media inventory and regeneration brief

`Asset-register.json` defines every distinct visual family across the approved public and dashboard boards, container usage, composition prompt, lighting/materials, crops, dimensions, formats, alt intent and authenticity status. Backgrounds and functional icons are native CSS/vector definitions, not AI raster assets. Shared imagery is intentionally reused between public category teaser and appropriate dashboard content thumbnail only where the underlying content really matches.

Generated marketing scenes are illustrative; real portfolio screenshots, real product previews, real print evidence and the real founder portrait must be acquired from approved user assets. Some mockups show invented clients/templates/portraits or physically questionable PC internals; those are not shipping assets. The only generated collection in this phase is the written prompt inventory, not images. Next image phase should generate the approved unique marketing scenes once, display them individually in chat and package originals plus crops/formats; real evidence is never generated to fill gaps.

Reusable photographic direction: navy/slate setting, black/white printed objects with plausible FDM layer texture, soft natural/studio light, restrained cyan accent, uncluttered workspace. Leave space for HTML text; no logos, watermark, pseudo-UI lettering or dimensional labels baked into photos. Brand words use original SVG and interface text. Precise measurement diagrams are native vector with validated geometry/labels, not generative pictures.

14. Edge cases and acceptance plan

`Acceptance-register.json` lists normal, empty, failure, retry, concurrent and unauthorized scenarios with priority, setup and measurable expected results. P0 means release-blocking confidentiality/financial correctness/data loss; P1 core workflow failure; P2 polish/optional module. This is a prioritized finite risk set, not a claim to predict all situations.

Minimum end-to-end flow on isolated test data: submit EN/UK/RU enquiry with accepted source photos -> inspect private inbox original -> link reviewed customer/company -> accepted quote revision -> service-specific job -> draft -> German-first bilingual issue -> authenticated saved PDF -> partial payment -> optional reviewed portfolio draft -> publish approved derivative -> verify public list/detail/media/sitemap. Also verify existing direct job-to-invoice path remains working, because quotes are an extension not a prerequisite to all invoices.

Use unit checks for decimals/status mapping/validators and PDF layout; integration checks for transaction/idempotency/permissions/private files; browser checks for original text, mobile menus/forms, retries and entire workflows; manually review long German and real-media layouts. Stub mail/merchant responses in tests; no unsolicited production mail or live invoice creation. Load/resource tests use bounded synthetic files in isolated environment. Verify restored backup records/files/checksums after recovery. Acceptance results are planned, not executed in this analysis.

15. Phased implementation and review gates

Phase 0 - reconcile sources and release baseline

Confirm working branch/deployed commit, runtime Node24 consistency and production DB binding. Preserve known tests and migrations. Establish clean isolated test environment, successful backup/restore evidence and baseline public/footer/mobile screenshots. Check relevant legal/tax/content availability and dependency/security status. No production reset. Exit: reproducible baseline and reviewed change inventory.

Phase 1 - design foundations and compatibility

Implement tokens, shared controls, accessible header/menu, Control shell, live footer preservation and responsive rules in an isolated development branch. Keep real content and old URLs. Add proposed module navigation only as functions become ready. Exit: visual comparison and keyboard/mobile tests with no broken current workflows.

Phase 2 - contact, data boundaries and larger uploads

Introduce scoped direct-upload pipeline with old small-upload compatibility until verified; metadata-only application requests, private derivatives, reservation cleanup, idempotent contact persistence, original Unicode text and robust inbox/SMTP states. Add enquiry links/private notes and duplicate review. Exit: max-size/failure/abuse/unauthorized cases pass; public write credentials remain least privilege.

Phase 3 - public three-service presentation and publication

Build approved pages using real projects, 3D category, authentic media and preserved footer. Extend SEO/routes/slug aliases and catalogue projection. Keep unavailable products honest. Exit: published-only content/media, locale metadata, existing projects and responsive tests pass.

Phase 4 - CRM, print specs, catalogue and jobs

Extend client profiles/current statuses; add service-specific details and tasks with additive migrations, decimal catalogue data, job attachments and estimates. Keep existing customer/job numbers and relationships. Exit: edits/version conflict, archive and quote-estimate-copy checks pass.

Phase 5 - German-first invoices and quotes

Change future renderer/layout version and tests while preserving original issued bytes. Keep numbering/row locks/payment checks and one-draft index. Add explicit EN translations, idempotency and correction proposal only after accounting review. Add quote revisions/acceptance/conversions. Exit: long-DE/EN pagination, snapshot immutability, concurrent issue/payment and email-state checks pass.

Phase 6 - products and optional operational modules

Publish actual Framer/models with verified merchant/license links. Add review moderation, receipts/reports, content editor, backups/MFA/reminders only where backing services exist. Exit: no fake integration buttons/data; publication/private boundaries verified.

Phase 7 - staged release and monitoring

On protected preview, run existing + new tests/typecheck/lint/build and documented browser/security checks. Review redirects/canonicals/sitemap, footer accuracy, auth sessions, database-role grants, file limits, backups, real contact delivery and DE-first invoice samples. Release approval applies to the concrete deployment change; no deployment is authorized by this analysis. Monitor errors/storage/rate-limits/mail outcomes, preserve rollback plan and recheck first real workflow.

16. Source register

Evidence files are pinned to the deployed commit rather than branch names that may move. Repository links are reference sources, not permissions to alter anything.

- E01: repository main and feature branch tree/README/package manifests; https://github.com/VladyslavTitov/tiladys_portfolio_website/tree/e82bff04aac9915585b10cc853fe31c558a5964b
- E02: read-only Vercel metadata for tiladys-public and tiladys-control production; commit e82bff04..., Node24, iad1, existing deployment protection. Tool observation 12 October 2026.
- E03: read-only Neon tiladys-database active main table/schema/index/migration metadata; Invoice/InvoiceLine/InvoicePayment and bilingual migration verified. No private invoice rows read.
- E04: deployed apps/control/lib/invoices.ts, invoice-pdf.ts, invoice-issuance-requirements.ts and invoice issue/PATCH/payment/PDF routes; packages/db/prisma/schema.prisma and unique-draft migration.
- E05: deployed apps/web/lib/contact-upload-limits.ts, contact-images.ts, contact-rate-limit.ts and app/api/contact/route.ts.
- E06: deployed public-projects-db.ts, public media route, authenticated contact-attachment routes and private file metadata.
- E07: deployed security.ts, trusted-origins.ts, rate-limit.ts, login route and mail/reply routes.
- E08: live https://tiladys.com/en footer screenshot observation and deployed Footer.tsx, SocialLinks.tsx, globals.css. Screenshot used only for inspection, not packaged.
- E09: deployed seo.ts, sitemap.ts, ProjectSlugAlias/public loader; live published public routes.
- S01: Vercel Functions limits and upload guidance: https://vercel.com/docs/functions/limitations ; https://vercel.com/kb/guide/how-to-bypass-vercel-body-size-limit-serverless-functions
- S02: Google localized versions: https://developers.google.com/search/docs/specialty/international/localized-versions
- S03: official federal e-invoice definition: https://e-rechnung-bund.de/e-rechnung/was-ist-eine-e-rechnung ; BMF FAQ and UStDV34a endpoints unavailable during this review, so business-specific tax/retention details remain unverified.
- S04: OWASP authorization: https://cheatsheetseries.owasp.org/cheatsheets/Authorization_Cheat_Sheet.html
- S05: OWASP CSRF: https://cheatsheetseries.owasp.org/cheatsheets/Cross-Site_Request_Forgery_Prevention_Cheat_Sheet.html
- S06: OWASP session management: https://cheatsheetseries.owasp.org/cheatsheets/Session_Management_Cheat_Sheet.html
- S07: OWASP uploads: https://cheatsheetseries.owasp.org/cheatsheets/File_Upload_Cheat_Sheet.html
- S08: OWASP SSRF: https://cheatsheetseries.owasp.org/cheatsheets/Server_Side_Request_Forgery_Prevention_Cheat_Sheet.html
- S09: GDPR primary text: https://eur-lex.europa.eu/eli/reg/2016/679/oj/eng ; privacy policy application requires business-specific review.
- S10: Google sitemap guidance: https://developers.google.com/search/docs/crawling-indexing/sitemaps/overview
- S11: Google noindex: https://developers.google.com/search/docs/crawling-indexing/block-indexing
- S12: Web Vitals: https://web.dev/articles/vitals

- S13: accessibility target WCAG2.2 AA plus proposed44px touch design: https://www.w3.org/TR/WCAG22/ . Target is a design goal, not a conformity claim.
- S14: Neon Postgres skill guidance reviewed for pooled runtime/direct migrations, branching and safe recovery planning; preserve the actual Prisma version and test its connection configuration. Provider plan-specific restrictions must be reconfirmed before applying changes.

17. Delivery boundary

This analysis is ready for review. The later Codex master prompt must be written only after review of this specification, and will reference these registers rather than assuming generated images are executable requirements. Later asset generation follows the approved asset list and authenticity constraints. The public website, Control application and current records remain unchanged.

Appendix A. Individual control inventory

The complete per-control contracts are in Control-register.json: validation, actor, permission, reads/writes, loading/disabled states, success, failure recovery, confirmation and related effects. This index provides concrete actions.

PUBLIC-SHELL

ID	Control	Concrete operation
C001	TiLADYS logo	Open the labelled public page or section
C002	Websites	Open the labelled public page or section
C003	3D Printing	Open the labelled public page or section
C004	PC Support	Open the labelled public page or section
C005	Portfolio	Open the labelled public page or section
C006	About	Open the labelled public page or section
C007	Contact	Open the labelled public page or section
C008	Start a project	Open the labelled public page or section
C009	Language	Open language choices; navigate to available equivalent page
C010	Menu / Close	Toggle top-aligned menu directly under header; Escape closes; restore trigger focus
C011	Skip to content	Move focus to main content

FOOTER

ID	Control	Concrete operation
C012	Logo	Open the labelled public page or section
C013	Home	Open the labelled public page or section
C014	Services	Open the labelled public page or section
C015	Portfolio	Open the labelled public page or section
C016	About	Open the labelled public page or section
C017	Contact	Open the labelled public page or section
C018	Email	Open the labelled public page or section
C019	Phone	Open the labelled public page or section
C020	Threads	Open the labelled public page or section
C021	WhatsApp	Open the labelled public page or section
C022	Telegram	Open the labelled public page or section
C023	Instagram	Open the labelled public page or section
C024	Terms of Use	Open the labelled public page or section
C025	Privacy Policy	Open the labelled public page or section
C026	Impressum	Open the labelled public page or section
C027	Contact Me	Open the labelled public page or section
C028	Open Contact Form	Open the labelled public page or section

P01

ID	Control	Concrete operation
C029	Get in touch	Open the labelled public page or section
C030	View services	Open the labelled public page or section
C031	Learn more about websites	Open the labelled public page or section
C032	Discover 3D printing	Open the labelled public page or section
C033	Get PC support	Open the labelled public page or section
C034	View portfolio	Open the labelled public page or section
C035	Selected project image / View project	Open the labelled public page or section
C036	Browse templates	Open the labelled public page or section
C037	Explore models	Open the labelled public page or section
C038	More about me	Open the labelled public page or section

P02

ID	Control	Concrete operation
C039	Open service category	Open the labelled public page or section
C040	Other digital services	Open the labelled public page or section
C041	Request quote	Open the labelled public page or section

P03

ID	Control	Concrete operation
C042	Custom websites / Request a website	Open the labelled public page or section
C043	Browse Framer templates	Open the labelled public page or section
C044	Business websites / Learn more	Open the labelled public page or section
C045	Landing pages / Learn more	Open the labelled public page or section
C046	Updates and maintenance / Learn more	Open the labelled public page or section
C047	View all projects	Open the labelled public page or section
C048	Gallery View project	Open the labelled public page or section
C049	Template preview	Open the labelled public page or section
C050	View on Framer	Open the labelled public page or section

P04

ID	Control	Concrete operation
C051	Request a custom print	Open the labelled public page or section
C052	Browse models	Open the labelled public page or section
C053	Print from your file	Open the labelled public page or section
C054	Discuss an idea	Open the labelled public page or section
C055	Small custom batches	Open the labelled public page or section
C056	View more in gallery	Open the labelled public page or section
C057	Print View project	Open the labelled public page or section
C058	View model	Open the labelled public page or section
C059	Other requirements / Ask me	Open the labelled public page or section

P05

ID	Control	Concrete operation
C060	Describe your issue	Open the labelled public page or section
C061	Explore services	Open the labelled public page or section
C062	Setup and software	Open the labelled public page or section
C063	Troubleshooting	Open the labelled public page or section
C064	Cleaning	Open the labelled public page or section
C065	SSD and RAM upgrades	Open the labelled public page or section
C066	Backup and data transfer	Open the labelled public page or section
C067	Remote support	Open the labelled public page or section
C068	Book a local appointment	Open the labelled public page or section

PUBLIC-SERVICES

ID	Control	Concrete operation
C069	FAQ question	Expand/collapse associated answer; keep heading focus

P06

ID	Control	Concrete operation
C070	All / Websites / 3D Printing / PC and IT / More	Filter published projects; retain older category codes
C071	Search projects	Search bounded public title/summary only
C072	Next / Previous page	Move cursor within published results
C073	Project card	Open the labelled public page or section
C074	Framer templates	Open the labelled public page or section
C075	3D models	Open the labelled public page or section
C076	Have a project in mind	Open the labelled public page or section

P07

ID	Control	Concrete operation
C077	Portfolio breadcrumb	Open the labelled public page or section
C078	Visit website	Open the labelled public page or section
C079	View GitHub	Open the labelled public page or section
C080	Request similar project	Open the labelled public page or section
C081	Related project	Open the labelled public page or section
C082	Select gallery thumbnail	Change/view published gallery image; labelled keyboard equivalent
C083	Previous image	Change/view published gallery image; labelled keyboard equivalent
C084	Next image	Change/view published gallery image; labelled keyboard equivalent
C085	Zoom image	Change/view published gallery image; labelled keyboard equivalent
C086	Close image viewer	Change/view published gallery image; labelled keyboard equivalent

P08

ID	Control	Concrete operation
C087	Framer / 3D models / category	Filter published available products
C088	Search products	Search public product text
C089	Load more	Load bounded published next page
C090	View product	Open the labelled public page or section

P09

ID	Control	Concrete operation
C091	Product breadcrumb	Open the labelled public page or section
C092	Preview	Open the labelled public page or section
C093	View on Framer / View model store	Open the labelled public page or section
C094	Request a print of this model	Open the labelled public page or section
C095	Gallery thumbnail	Select actual product preview

P10

ID	Control	Concrete operation
C096	Let us work together / Get in touch	Open the labelled public page or section
C097	Local / online support enquiry	Open the labelled public page or section

P11

ID	Control	Concrete operation
C098	Topic	Edit draft value
C099	Request mode	Edit draft value
C100	Name	Edit draft value
C101	Email	Edit draft value
C102	Phone optional	Edit draft value
C103	Message	Edit draft value
C104	Preferred reply language optional	Edit draft value
C105	Privacy notice acknowledgement	Edit draft value
C106	Add photos / Choose files / Drop files	Reserve/upload private source photos with progress; finalize only approved files
C107	Remove selected photo	Remove local selection or unclaimed reservation; adjust source aggregate
C108	Retry failed upload	Retry/resume same reservation with bounded cost and unchanged form
C109	Attach model (conditional)	Upload private STL only after separate pipeline enabled
C110	Send request	Validate/finalize and persist enquiry once; return receipt
C111	Submit another request	Clear only after successful receipt and deliberate action
C112	Send Email	Open the labelled public page or section
C113	Call Now	Open the labelled public page or section
C114	View on map	Open the labelled public page or section
C115	Privacy Policy	Open the labelled public page or section

P11-REVIEW

ID	Control	Concrete operation
C116	Display name	Edit draft value
C117	Review text	Edit draft value
C118	Rating optional	Edit draft value
C119	Publication permission	Edit draft value
C120	Add review photos	Same protected source photo pipeline
C121	Submit review	Persist pending moderation record, never public automatically

P12

ID	Control	Concrete operation
C122	Legal table of contents	Open the labelled public page or section
C123	Home / Services / Contact on 404	Open the labelled public page or section
C124	Clear filters	Reset current search/filter without hiding load errors
C125	Retry content	Re-read same published safe data request

ADMIN-SHELL

ID	Control	Concrete operation
C126	Overview	Open authorized section; show unavailable future module honestly
C127	Inbox	Open authorized section; show unavailable future module honestly
C128	Clients	Open authorized section; show unavailable future module honestly
C129	Jobs	Open authorized section; show unavailable future module honestly
C130	Quotes and Invoices	Open authorized section; show unavailable future module honestly
C131	Portfolio	Open authorized section; show unavailable future module honestly
C132	Products	Open authorized section; show unavailable future module honestly
C133	Tasks	Open authorized section; show unavailable future module honestly
C134	Settings	Open authorized section; show unavailable future module honestly
C135	Reviews	Open authorized section; show unavailable future module honestly
C136	Reports	Open authorized section; show unavailable future module honestly
C137	Global search	Search allowed customers/jobs/invoices/projects with bounded paged results; type filter
C138	Notifications	Open actual due/failure/unread records, not invented events
C139	Interface language	Change UI language; preserve original enquiry and invoice order
C140	Collapse sidebar / Mobile menu / Close	Toggle labelled navigation; keyboard/focus/scroll treatment
C141	View public website	Open public homepage in separate safe tab
C142	Profile / Account	Open account settings
C143	Log out	Revoke current server session and clear cookie

D01

ID	Control	Concrete operation
C144	New enquiries	Navigate to specific authorized filtered page or creation draft
C145	Active jobs	Navigate to specific authorized filtered page or creation draft
C146	Awaiting payment	Navigate to specific authorized filtered page or creation draft
C147	Tasks due today	Navigate to specific authorized filtered page or creation draft
C148	New client	Navigate to specific authorized filtered page or creation draft
C149	Create quote	Navigate to specific authorized filtered page or creation draft
C150	New job	Navigate to specific authorized filtered page or creation draft
C151	View all jobs	Navigate to specific authorized filtered page or creation draft
C152	View all enquiries	Navigate to specific authorized filtered page or creation draft
C153	Quick access Portfolio	Navigate to specific authorized filtered page or creation draft
C154	Quick access Framer / models	Navigate to specific authorized filtered page or creation draft
C155	Open job / enquiry / invoice row	Navigate to specific authorized filtered page or creation draft
C156	Complete Today task	Persist reversible task completion

D02

ID	Control	Concrete operation
C157	All / Unread / Archived / Spam	Filter private inbox
C158	Search messages	Bounded search in authorized original messages
C159	Next / Previous	Page private inbox
C160	Select enquiry	Open detail and explicitly mark unread as read once
C161	Preview / Download attachment	Authorize message-attachment relationship; safe private response
C162	Internal note	Edit draft value
C163	Reply message	Edit draft value
C164	Save note	Persist a private author/time-linked note
C165	Link client	Review suggested duplicate and bind selected existing client
C166	Create client	Open reviewed prefilled client form, then allocate once
C167	Create job	Open linked source brief as a private draft
C168	Create quote	Open linked private quote draft
C169	Send reply	Send using configured SMTP; record accepted/failed/unknown delivery honestly
C170	Archive / Restore enquiry	Reversible state change
C171	Mark spam / Restore	Reversible moderation flag
C172	Reply by email client (optional)	Open mailto; do not mark delivered automatically

D03

ID	Control	Concrete operation
C173	All / Individuals / Companies	Filter type projection without destroying underlying Customer/Company
C174	Client search	Bounded number/name/email/company query
C175	Profile Overview / Jobs / Quotes / Invoices / Notes / Files / Activity	Select authorized linked panel
C176	Client status	Filter LEAD/ACTIVE/INACTIVE/ARCHIVED
C177	New client	Open empty client draft
C178	Open client row	Read authorized client profile
C179	Save client / Edit	Validate and persist reviewed fields; allocate only on create
C180	Create job	Open customer-linked job draft
C181	Create quote	Open linked quote draft
C182	Add note / Save note	Persist private author-linked note
C183	Archive / Restore client	Reversible state with financial references preserved
C184	Export clients	Create scoped private sanitized export and audit
C185	Individual / Company type	Edit draft value
C186	Company name / linked company	Edit draft value
C187	First name / Last name / contact name	Edit draft value
C188	Email	Edit draft value
C189	Phone / secondary phone	Edit draft value
C190	Street / Postcode / City / Country	Edit draft value
C191	Preferred language	Edit draft value
C192	Source	Edit draft value
C193	Private notes	Edit draft value

D04

ID	Control	Concrete operation
C194	All services / Websites / 3D / PC	Filter existing serviceType
C195	Board / List	Switch same data presentation
C196	Job status / dates / client	Bounded current enum/date query
C197	Detail Details / Tasks / Files / Activity	Open selected job panel
C198	New job	Open job creation draft
C199	Save draft job / Save changes	Validate relation and service-specific fields; detect stale version
C200	Move stage	Change authorized production stage with keyboard alternative
C201	Mark ready	Record production readiness; no auto billing
C202	Complete / Cancel job	Reviewed business transition preserving history
C203	Open quote	Read linked quote revision
C204	Create invoice	Reuse existing active draft or create one explicit linked draft
C205	Add task	Open job-linked task draft
C206	Create portfolio draft	Select publishable copy/media and strip private details
C207	Upload job file	Private scoped file flow
C208	Download job file	Authorize association and private download
C209	Remove file association	Reviewed association removal/retention decision
C210	Linked client / company	Edit draft value
C211	Job title	Edit draft value
C212	Service category	Edit draft value
C213	Due/service date and time	Edit draft value
C214	Status	Edit draft value
C215	Description / private notes / customer-visible notes	Edit draft value
C216	Website brief	Edit draft value
C217	PC device details	Edit draft value
C218	Print dimensions L/W/H and units	Edit draft value
C219	Print quantity / material / colour	Edit draft value

ID	Control	Concrete operation
C220	Selected model	Edit draft value
C221	Print estimate hours / grams	Edit draft value
C222	Checklist item	Edit draft value
C223	Catalogue service picker	Edit draft value
C224	Quantity / unit / agreed unit price	Edit draft value
C225	Tax treatment / rate / billing period / price confirmed	Edit draft value
C226	Internal unit cost	Edit draft value

D05

ID	Control	Concrete operation
C227	Quotes / Invoices	Switch private resource list
C228	Draft / Sent / Accepted / Declined / Expired	Filter quote status
C229	Search quote / client / date	Bounded private search
C230	New quote	Create linked private draft with fresh conversion key
C231	Save quote draft	Validate snapshots and draft version
C232	Preview quote	Render draft without sending/accepting
C233	Send quote	Final review and tracked SMTP acceptance; freeze sent revision
C234	Record acceptance	Record exact revision/date/source evidence
C235	Record decline / expire	Reviewed status with reason
C236	Revise sent quote	Create new revision preserving old agreed text
C237	Convert accepted quote to job	Idempotently create or return linked job
C238	Create invoice from accepted quote	Reviewed idempotent draft from accepted snapshot
C239	Client / recipient	Edit draft value
C240	Valid until	Edit draft value
C241	Service items / qty / unit / price / tax	Edit draft value
C242	Scope / exclusions / notes EN and DE	Edit draft value
C243	Acceptance source / date / reference	Edit draft value

D06

ID	Control	Concrete operation
C244	Invoice search / date	Bounded number/client/date query
C245	Document state	DRAFT/ISSUED/CANCELLED
C246	Payment state / Overdue	Derived UNPAID/PARTIAL/PAID and due positive balance
C247	Create invoice	Open empty/linked draft; allocate legal number only on issue
C248	From accepted quote / job	Return unique active linked draft or create reviewed one
C249	Save draft	Validate translated fields, recipient, lines and version under lock
C250	Preview PDF	Render draft DE first EN second; no number allocation
C251	Issue invoice	Review missing requirements then lock/validate/freeze/allocate exactly once
C252	Customer / job links	Edit draft value
C253	Individual / Company billing recipient	Edit draft value
C254	Recipient name / company / address / email	Edit draft value
C255	Invoice / due / service dates	Edit draft value
C256	Service name EN / DE	Edit draft value
C257	Description EN / DE and unit EN / DE	Edit draft value
C258	Catalogue search	Edit draft value
C259	Line quantity / price / price mode	Edit draft value
C260	Tax treatment	Edit draft value
C261	Notes EN / DE	Edit draft value
C262	German + English PDF (fixed)	Edit draft value

D07

ID	Control	Concrete operation
C263	Download PDF / Preview issued PDF	Return authorized stored original bytes; no historical rerender
C264	Record payment	Lock invoice, validate positive amount<=remaining, idempotent record
C265	Save payment	Persist validated manual payment and return true balance
C266	Create correction / Cancel invoice	Reviewed compliant linked correction preserving original snapshot
C267	View payment history / Activity	Read ledger and document transitions
C268	Payment amount	Edit draft value
C269	Payment date	Edit draft value
C270	Payment method / reference / note	Edit draft value
C271	Correction reason	Edit draft value

D08

ID	Control	Concrete operation
C272	Project category / status	Filter existing category and DRAFT/PUBLISHED/ARCHIVED
C273	Search / display order	Bounded project search and safe order
C274	EN / DE / UK / RU / SK / FR	Edit chosen localized field, preserve other keys
C275	New project	Open project draft
C276	Edit project	Read authorized editable project
C277	Save draft	Persist validated fields/translations/media selection
C278	Preview public portfolio / project	Authenticated non-indexable preview, actual published route separate
C279	Publish project	Review public text/media/rights then publish and invalidate projections
C280	Archive / Unpublish / Restore draft	Reversible status, immediate public hide and media checks
C281	Add gallery images	Upload approved owner portfolio images with scoped limit
C282	Remove gallery image	Review association removal, preserve unrelated files
C283	Reorder / choose cover	Save explicit media order/cover belonging to project
C284	Create from completed job	Reviewed sanitized draft; no client info automatically
C285	Title / type / summary / description / role / work list	Edit draft value
C286	Slug	Edit draft value
C287	Category	Edit draft value
C288	Status	Edit draft value
C289	Display order / curated homepage selection	Edit draft value
C290	Completion date / tools / website / GitHub	Edit draft value
C291	SEO title / description / social image	Edit draft value
C292	Cover image / localized alt	Edit draft value

D09

ID	Control	Concrete operation
C293	All / Framer templates / 3D models	Filter Product type
C294	Product search / state / sort	Bounded published/private listing query
C295	New product	Open draft listing
C296	Save draft product	Validate actual listing/links/license/translation fields
C297	Add product images	Owner-scoped actual preview assets
C298	Preview product	Private non-indexable preview
C299	Publish product	Verify actual availability/license/destination; explicit release
C300	Archive product	Hide listing/purchase action, preserve aliases/history
C301	Open merchant / demo	Open validated URL, no inferred synchronization
C302	Product name / type / slug	Edit draft value
C303	Description / included files / formats	Edit draft value
C304	Price / currency / availability	Edit draft value
C305	License summary	Edit draft value
C306	Preview / external purchase URL	Edit draft value
C307	Visibility and language tabs	Edit draft value

D10

ID	Control	Concrete operation
C308	Today / Upcoming / Completed	Filter authoritative due/state
C309	List / Calendar	Render same records
C310	Priority / linked job / date range	Bounded task filters
C311	Add task	Open linked private draft
C312	Save task / Edit	Validate title/due/links/version
C313	Mark complete / Undo	Reversible persisted completion
C314	Set reminder (conditional)	Schedule only when real delivery service exists
C315	Title / note	Edit draft value
C316	Linked client/job	Edit draft value
C317	Due date/time and zone	Edit draft value
C318	Priority / state	Edit draft value
C319	Reminder optional	Edit draft value

D11

ID	Control	Concrete operation
C320	Websites / 3D / PC / section	Filter catalogue without rewriting old categories
C321	Search services	Code plus actual localized text
C322	Language tabs	Edit actual translation, keep other locale values
C323	New service / section	Create code-unique draft row
C324	Save service / all changes	Atomic validated moderate bulk update; row errors identified
C325	Duplicate service	New unique code, explicit review
C326	Archive / deactivate service	Retain historic snapshots; hide future picker default
C327	Calculate print estimate	Compute explicit costs/markup or margin; no mutation
C328	Use estimate in draft quote	Copy reviewed amounts once, configured tax separate
C329	Code / category / display order / active	Edit draft value
C330	Name / note / unit / price mode	Edit draft value
C331	Default numeric price / display string	Edit draft value
C332	Filament grams / price per kg	Edit draft value
C333	Print hours / machine rate / labour hours / rate	Edit draft value
C334	Fixed costs / batch quantity / allowance	Edit draft value
C335	Markup % / target margin %	Edit draft value

D12

ID	Control	Concrete operation
C336	Business / Invoices / Website / Languages / Uploads / Integrations	Open selected settings section
C337	Save settings	Validate owner-confirmed business defaults; future-document effect only
C338	Upload logo / portrait	Trusted owner asset, preserve original logo; actual face only
C339	Preview desktop / mobile	Render unpublished settings/content without public mutation
C340	Publish website copy	Review schema/translations and public facts
C341	Connect integration (conditional)	Supported scoped service flow only, no fake connected state
C342	Business name / public email / phone / address	Edit draft value
C343	Invoice/client/job numbering	Edit draft value
C344	Currency / timezone / payment terms	Edit draft value
C345	Tax mode / identifier / statements DE and EN	Edit draft value
C346	Bank account holder / IBAN / BIC / bank name	Edit draft value
C347	Payment instructions DE and EN	Edit draft value
C348	Languages	Edit draft value
C349	Upload policy	Edit draft value
C350	About biography / portrait crop	Edit draft value
C351	Integration settings	Edit draft value

D13

ID	Control	Concrete operation
C352	Login email / password / existing secret phrase	Edit draft value
C353	Show/hide credential	Edit draft value
C354	Two-factor code (conditional)	Edit draft value
C355	Recovery token (conditional)	Edit draft value
C356	Sign in	Rate-limited Argon2 verification; fresh opaque session
C357	Forgot password (conditional)	Generic response; actual expiring recovery flow only
C358	Change password	Reauthenticate; updated hash and session policy
C359	Set up MFA (conditional)	Verify real TOTP/WebAuthn; recovery-code review
C360	Revoke session	Owner-authorized selected session termination
C361	Session expired / Back to sign in	Return to login; preserve only safe nonfinancial draft policy; recheck next mutation

D14

ID	Control	Concrete operation
C362	Export clients/jobs/invoices CSV	Scoped private export, neutralize formula injection and audit
C363	Export uploaded files	Authorized aggregate archive with safe paths; protected expiry
C364	Download export	Authorize current owner and expiry; no public URL
C365	Configure backup (conditional)	Actual backup policy/provider setup; no false success
C366	Review restore	Display source/time/environment and plan; no autonomous overwrite
C367	Account / Data / Activity	Select authorized settings panel
C368	Activity date / action / entity	Bounded audit filters

D15

ID	Control	Concrete operation
C369	Approve and publish review	Check consent/media/actual text; explicit moderation
C370	Keep pending	Persist pending review without publication
C371	Archive / reject review	Reversible moderation with reason
C372	Open website content page	Read corresponding editor, not auto publish
C373	Pending / Published / Archived	Filter Review state
C374	Select review	Read original text/media permission
C375	Content language / page	Open chosen versioned content editor

D16

ID	Control	Concrete operation
C376	Add expense / Save expense	Validate actual decimal amount/receipt/category and source
C377	Attach receipt	Private scoped upload; safe type/size
C378	Add payment	Open linked invoice payment action, not fake bank-sync
C379	Export report	Scoped operational-period private export
C380	Report date range / service / currency	Select coherent source period/category/currency
C381	Payments / Expenses	Read selected operational ledger
C382	Expense description / date / amount / currency	Edit draft value
C383	Category / linked job / receipt	Edit draft value

SHARED-STATES

ID	Control	Concrete operation
C384	Retry	Retry same bounded read/action only after known failure
C385	Clear search	Reset filter query and cursor
C386	Close drawer / Back	Return to list and restore originating focus
C387	Save / Discard / Stay on page	Resolve unsaved draft explicitly
C388	Cancel confirmation	Close without mutation
C389	Confirm archive	Reversible authorized archival, never irreversible data purge
C390	Upload cancel / retry	Bounded same reservation cleanup/retry

Appendix B. Unique asset prompts and acquisition briefs

Asset-register.json also contains lighting, materials, negative space, reuse restrictions and delivery policies. Shared illustration direction is navy/slate, restrained cyan, black/white PLA, physically plausible geometry and real lighting. Keep copy and measurements in native HTML/SVG.

B01 - Original wordmark

Source: original-vector. Containers: public-header, live-footer, control-sidebar, login, invoice.

Acquire original /brand/icon.svg from deployed repo. Preserve vector geometry, white shape/cyan dot and proportions; do not ask imagegen to recreate logo. Exact original asset; no pseudo-logo or generated brand lettering.

Master: Original SVG; trusted derivatives only; ratio original445:106. Mobile: Retain original vector proportions and responsive display size; do not crop the brand mark..

Alt: TiLADYS.

B02 - Circuit logo mark

Source: original-vector. Containers: footer-brand, footer-legal.

Acquire original /brand/logo.svg. Use decorative 78x58 and 118x88 displays matching live footer; invoice use only when requested.

Master: Original SVG; ratio original. Mobile: Retain original vector proportions and responsive display size; do not crop the brand mark..

Alt: Decorative empty alt.

BG01 - Public hero and CTA backgrounds

Source: native-css. Containers: all-public-heroes, contact-CTA.

Native CSS navy #071837 with restrained tone. Real text/buttons overlay; solid/controlled subtle background; avoid raster replacement. No generated text or per-page giant background downloads.

Master: No raster; ratio 16:9. Mobile: Not applicable: use responsive native CSS, SVG or data layout; do not rasterize text or controls..

Alt: Decorative alt empty when adjacent text supplies purpose; actual work/product images have specific localized descriptive alt.

BG02 - Live footer background

Source: native-css. Containers: all-public-footers.

Retain native radial and linear gradient from deployed globals.css. Radial blue accent 92%/38%; linear 110deg #001025,#041d3e,#00162f; preserve native layout. Latest user-approved footer overrides every generated one-row footer.

Master: No raster; ratio 16:9. Mobile: Not applicable: use responsive native CSS, SVG or data layout; do not rasterize text or controls..

Alt: Decorative alt empty when adjacent text supplies purpose; actual work/product images have specific localized descriptive alt.

BG03 - Dashboard surfaces

Source: native-css. Containers: control-canvas, cards, forms, sidebar.

Native CSS canvas #F4F7FA, white cards, navy sidebar, border #DBE4EE. Spacing/radius tokens; no desktop wallpaper or photographic admin background.

Master: No raster; ratio 16:9. Mobile: Not applicable: use responsive native CSS, SVG or data layout; do not rasterize text or controls..

Alt: Decorative alt empty when adjacent text supplies purpose; actual work/product images have specific localized descriptive alt.

IC01 - Functional icon set

Source: native-vector. Containers: service-cards, forms, process, footer, control-actions.

Reuse Lucide icons and original existing social SVG. Consistent 18-24px line icons; decorative icon hidden when text names action; accessible label for icon-only controls. Do not generate icons or replace working branded social identifiers with random symbols.

Master: SVG; ratio 1:1. Mobile: Not applicable: use responsive native CSS, SVG or data layout; do not rasterize text or controls..

Alt: Decorative alt empty when adjacent text supplies purpose; actual work/product images have specific localized descriptive alt.

G01 - Three-service home hero

Source: generate-illustration. Containers: P01.hero, P01.service-teasers.

Photorealistic professional TiLADYS marketing illustration, a tidy widescreen monitor, a restrained correctly assembled black PC tower and practical black/white PLA prints. Eye-level three-quarter desktop view, subjects grouped in right 55%; monitor behind organizer and generic small printed boat, tower right; no excessive RGB, plants only small visual balance Soft studio/daylight from upper left; gentle fill, controlled shadows; no neon glow. Materials: realistic visible fine FDM layers, matte charcoal chassis, subtle brushed desk, blank clean monitor for actual preview insertion. Brand palette navy #071837, restrained cyan #7be0df, white and charcoal. Clean composition, consistent colour grading, realistic scale and perspective. No logos, watermarks, baked interface copy or dimensional labels.

Master: 2400x1350 master; responsive 1600/1200/800/480 derivatives; ratio 16:9. Mobile: Keep critical object inside central 65%; do not crop screen/hand/contact point; use separate 4:3 crop or stack whole illustration below text.

Alt: Illustration of website creation, 3D printing and PC support.

G02 - Website service hero

Source: generate-illustration. Containers: P03.hero, P01.website-card.

Photorealistic professional TiLADYS marketing illustration, a slim laptop or desktop monitor beside a phone showing blank neutral screen areas for real site-preview insertion. Three-quarter view, larger display back-left within right-side group, phone front-right; careful perspective and natural device proportions Soft studio/daylight from upper left; gentle fill, controlled shadows; no neon glow. Materials: matte black/glass device frames, neutral blue screen placeholder, discreet desk. Brand palette navy #071837, restrained cyan #7be0df, white and charcoal. Clean composition, consistent colour grading, realistic scale and perspective. No logos, watermarks, baked interface copy or dimensional labels.

Master: 2400x1350 master; responsive 1600/1200/800/480 derivatives; ratio 16:9. Mobile: Keep critical object inside central 65%; do not crop screen/hand/contact point; use separate 4:3 crop or stack whole illustration below text.

Alt: Decorative alt empty when adjacent text supplies purpose; actual work/product images have specific localized descriptive alt.

G03 - 3D printing collection hero

Source: generate-illustration. Containers: P04.hero, P01.print-card.

Photorealistic professional TiLADYS marketing illustration, a matte black compartment desk organizer, white tray, faceted white planter and white angled phone stand. Product group on navy tabletop, organizer rear-left, planter rear-right, stand front-right, tray foreground; authentic small-printer scale Soft studio/daylight from upper left; gentle fill, controlled shadows; no neon glow. Materials: plausible PLA FDM layers, matte black/white, no injection-moulded gloss. Brand palette navy #071837, restrained cyan #7be0df, white and charcoal. Clean composition, consistent colour grading, realistic scale and perspective. No logos, watermarks, baked interface copy or dimensional labels.

Master: 2400x1350 master; responsive 1600/1200/800/480 derivatives; ratio 16:9. Mobile: Keep critical object inside central 65%; do not crop screen/hand/contact point; use separate 4:3 crop or stack whole illustration below text.

Alt: Illustrative black and white PLA objects.

G04 - PC support hero

Source: generate-illustration. Containers: P05.hero, P01.pc-card.

Photorealistic professional TiLADYS marketing illustration, a realistic black desktop computer with open side, installed fan cooler/RAM/GPU and neatly routed cables. Tower at right on wood/slate workbench, modest tools laid foreground; visible correctly attached hardware, no invented circuit architecture Soft studio/daylight from upper left; gentle fill, controlled shadows; no neon glow. Materials: steel mesh, real motherboard topology, restrained blue fan light, matte bench. Brand palette navy #071837, restrained cyan #7be0df, white and charcoal. Clean composition, consistent colour grading, realistic scale and perspective. No logos, watermarks, baked interface copy or dimensional labels. No unsafe assembly, loose liquid, impossible connectors or fictitious GPU lettering.

Master: 2400x1350 master; responsive 1600/1200/800/480 derivatives; ratio 16:9. Mobile: Keep critical object inside central 65%; do not crop screen/hand/contact point; use separate 4:3 crop or stack whole illustration below text.

Alt: Decorative alt empty when adjacent text supplies purpose; actual work/product images have specific localized descriptive alt.

G05 - Portfolio hero

Source: generate-illustration. Containers: P06.hero.

Photorealistic professional TiLADYS marketing illustration, a monitor with a blank preview insert, simple white printed pen cup/boat and small plant. Calm right-weighted composition with large negative space left; symbolically digital and physical work, not client proof Soft studio/daylight from upper left; gentle fill, controlled shadows; no neon glow. Materials: fine-layer PLA, matte monitor, navy desk. Brand palette navy #071837, restrained cyan #7be0df, white and charcoal. Clean composition, consistent colour grading, realistic scale and perspective. No logos, watermarks, baked interface copy or dimensional labels.

Master: 2400x1350 master; responsive 1600/1200/800/480 derivatives; ratio 16:9. Mobile: Keep critical object inside central 65%; do not crop screen/hand/contact point; use separate 4:3 crop or stack whole illustration below text.

Alt: Decorative alt empty when adjacent text supplies purpose; actual work/product images have specific localized descriptive alt.

G06 - About workbench hero

Source: generate-illustration. Containers: P10.hero.

Photorealistic professional TiLADYS marketing illustration, a modest desk with task lamp, closed laptop, pen cup and two white printed objects. Eye-level warm workplace, grouped at right; lamp bends over laptop; small plant/pegboard optional, left remains navy Soft studio/daylight from upper left; gentle fill, controlled shadows; no neon glow. Materials: black lamp, wood/slate surface, ordinary paper and layered PLA. Brand palette navy #071837, restrained cyan #7be0df, white and charcoal. Clean composition, consistent colour grading, realistic scale and perspective. No logos, watermarks, baked interface copy or dimensional labels. This is illustrative workspace, not a claim of an actual TiLADYS office.

Master: 2400x1350 master; responsive 1600/1200/800/480 derivatives; ratio 16:9. Mobile: Keep critical object inside central 65%; do not crop screen/hand/contact point; use separate 4:3 crop or stack whole illustration below text.

Alt: Decorative alt empty when adjacent text supplies purpose; actual work/product images have specific localized descriptive alt.

G07 - Contact workbench hero

Source: generate-illustration. Containers: P11.hero.

Photorealistic professional TiLADYS marketing illustration, an inviting clear desk with closed laptop, pen holder and small plant. Right-grouped calm workspace with chair edge, no readable slogan on wall; left blank for HTML contact headline Soft studio/daylight from upper left; gentle fill, controlled shadows; no neon glow. Materials: muted wood, navy wall, matte hardware. Brand palette navy #071837, restrained cyan #7be0df, white and charcoal. Clean composition, consistent colour grading, realistic scale and perspective. No logos, watermarks, baked interface copy or dimensional labels. No wall poster with baked copy; no fictitious office location or contact information.

Master: 2400x1350 master; responsive 1600/1200/800/480 derivatives; ratio 16:9. Mobile: Keep critical object inside central 65%; do not crop screen/hand/contact point; use separate 4:3 crop or stack whole illustration below text.

Alt: Decorative alt empty when adjacent text supplies purpose; actual work/product images have specific localized descriptive alt.

G08 - Optional legal-page illustration

Source: generate-illustration. Containers: P12.legal-hero.

Photorealistic professional TiLADYS marketing illustration, subtle brass balance scale and neutral books on a navy surface. Small restrained right-side still life, no symbols implying legal certification; optional, default legal header is text-only Soft studio/daylight from upper left; gentle fill, controlled shadows; no neon glow. Materials: brushed brass, blank-cover books. Brand palette navy #071837, restrained cyan #7be0df, white and charcoal. Clean composition, consistent colour grading, realistic scale and perspective. No logos, watermarks, baked interface copy or dimensional labels. Optional decorative asset only; no generated legal text, firm insignia or certification.

Master: 2400x1350 master; responsive 1600/1200/800/480 derivatives; ratio 16:9. Mobile: Keep critical object inside central 65%; do not crop screen/hand/contact point; use separate 4:3 crop or stack whole illustration below text.

Alt: Decorative alt empty when adjacent text supplies purpose; actual work/product images have specific localized descriptive alt.

G09 - Compartment desk organizer concept

Source: generate-illustration. Containers: P04.gallery, P07.example, D08.example, D09.example.

Photorealistic professional TiLADYS marketing illustration, a black PLA desk organizer with several open rectangular compartments holding ordinary pens/scissors. Centered three-quarter product photo on quiet desk, whole object visible, no hands; edge structure plausibly printable Soft studio/daylight from upper left; gentle fill, controlled shadows; no neon glow. Materials: matte black layered PLA, clean but not falsely perfect surfaces. Brand palette navy #071837, restrained cyan #7be0df, white and charcoal. Clean composition, consistent colour grading, realistic scale and perspective. No logos, watermarks, baked interface copy or dimensional labels.

Master: 2000x1500; ratio 4:3. Mobile: Keep critical object inside central 65%; do not crop screen/hand/contact point; use separate 4:3 crop or stack whole illustration below text.

Alt: Concept illustration of a black printed desk organizer.

G10 - Faceted white planter concept

Source: generate-illustration. Containers: P04.gallery, P06.example, D09.example.

Photorealistic professional TiLADYS marketing illustration, a white faceted planter with small green plant. Centered three-quarter composition, neutral wood/slate table, soft background, crop contains full planter lip/base Soft studio/daylight from upper left; gentle fill, controlled shadows; no neon glow. Materials: white layered PLA, clear planar facets, soil hidden naturally. Brand palette navy #071837, restrained cyan #7be0df, white and charcoal. Clean composition, consistent colour grading, realistic scale and perspective. No logos, watermarks, baked interface copy or dimensional labels.

Master: 2000x1500; ratio 4:3. Mobile: Keep critical object inside central 65%; do not crop screen/hand/contact point; use separate 4:3 crop or stack whole illustration below text.

Alt: Decorative alt empty when adjacent text supplies purpose; actual work/product images have specific localized descriptive alt.

G11 - White phone stand concept

Source: generate-illustration. Containers: P04.gallery, P09.example, D06.example.

Photorealistic professional TiLADYS marketing illustration, a minimal white angled PLA phone stand supporting a plain black phone. Three-quarter side view shows base and front lip, realistic centre of gravity, blank phone screen Soft studio/daylight from upper left; gentle fill, controlled shadows; no neon glow. Materials: matte white fine layers, black glass phone. Brand palette navy #071837, restrained cyan #7be0df, white and charcoal. Clean composition, consistent colour grading, realistic scale and perspective. No logos, watermarks, baked interface copy or dimensional labels.

Master: 2000x1500; ratio 4:3. Mobile: Keep critical object inside central 65%; do not crop screen/hand/contact point; use separate 4:3 crop or stack whole illustration below text.

Alt: Decorative alt empty when adjacent text supplies purpose; actual work/product images have specific localized descriptive alt.

G12 - Black phone stand concept

Source: generate-illustration. Containers: P01.print-teaser, D06.example, D09.example.

Photorealistic professional TiLADYS marketing illustration, a black ribbed minimalist phone stand without or with a plain phone. Centered front three-quarter product view; whole base/support visible; simple feasible geometry Soft studio/daylight from upper left; gentle fill, controlled shadows; no neon glow. Materials: charcoal PLA with modest ridges, no glossy impossible fins. Brand palette navy #071837, restrained cyan #7be0df, white and charcoal. Clean composition, consistent colour grading, realistic scale and perspective. No logos, watermarks, baked interface copy or dimensional labels.

Master: 2000x1500; ratio 4:3. Mobile: Keep critical object inside central 65%; do not crop screen/hand/contact point; use separate 4:3 crop or stack whole illustration below text.

Alt: Decorative alt empty when adjacent text supplies purpose; actual work/product images have specific localized descriptive alt.

G13 - Cable clip concept

Source: generate-illustration. Containers: P04.gallery, P04.model-card.

Photorealistic professional TiLADYS marketing illustration, small black and white PLA clips securing an ordinary cable. Macro three-quarter desk photo, clip openings mechanically plausible; separate clean single-object variant for model card Soft studio/daylight from upper left; gentle fill, controlled shadows; no neon glow. Materials: layered PLA, real cable bend and shadows. Brand palette navy #071837, restrained cyan #7be0df, white and charcoal. Clean composition, consistent colour grading, realistic scale and perspective. No logos, watermarks, baked interface copy or dimensional labels.

Master: 2000x1500; ratio 4:3. Mobile: Keep critical object inside central 65%; do not crop screen/hand/contact point; use separate 4:3 crop or stack whole illustration below text.

Alt: Decorative alt empty when adjacent text supplies purpose; actual work/product images have specific localized descriptive alt.

G14 - Electronics enclosure concept

Source: generate-illustration. Containers: P04.gallery, P07.example.

Photorealistic professional TiLADYS marketing illustration, a compact white PLA rectangular electronics enclosure with ventilation slots and screw holes. Three-quarter clean tabletop view, enclosure whole and proportionate; optional lid shown slightly separated Soft studio/daylight from upper left; gentle fill, controlled shadows; no neon glow. Materials: matte layered PLA, plausible wall thickness/slots. Brand palette navy #071837, restrained cyan #7be0df, white and charcoal. Clean composition, consistent colour grading, realistic scale and perspective. No logos, watermarks, baked interface copy or dimensional labels. No claim this is an actual certified safety enclosure or electrical product.

Master: 2000x1500; ratio 4:3. Mobile: Keep critical object inside central 65%; do not crop screen/hand/contact point; use separate 4:3 crop or stack whole illustration below text.

Alt: Decorative alt empty when adjacent text supplies purpose; actual work/product images have specific localized descriptive alt.

G15 - Hex-pattern pen cup concept

Source: generate-illustration. Containers: P04.model-card, P01.print-card.

Photorealistic professional TiLADYS marketing illustration, a white cylindrical PLA pen cup with honeycomb surface pattern. Clean product-only three-quarter crop, intact printable cell walls and flat base; optional

pens separate Soft studio/daylight from upper left; gentle fill, controlled shadows; no neon glow. Materials: visible FDM layers and realistic wall geometry. Brand palette navy #071837, restrained cyan #7be0df, white and charcoal. Clean composition, consistent colour grading, realistic scale and perspective. No logos, watermarks, baked interface copy or dimensional labels.

Master: 2000x1500; ratio 4:3. Mobile: Keep critical object inside central 65%; do not crop screen/hand/contact point; use separate 4:3 crop or stack whole illustration below text.

Alt: Decorative alt empty when adjacent text supplies purpose; actual work/product images have specific localized descriptive alt.

G16 - Decorative gear concept

Source: generate-illustration. Containers: P01.model-teaser, P07.related.

Photorealistic professional TiLADYS marketing illustration, a small matte gray or black PLA decorative gear. Centered three-quarter view, plausible consistent tooth pattern; not presented as a precision drive component Soft studio/daylight from upper left; gentle fill, controlled shadows; no neon glow. Materials: layered PLA, subtle shadows. Brand palette navy #071837, restrained cyan #7be0df, white and charcoal. Clean composition, consistent colour grading, realistic scale and perspective. No logos, watermarks, baked interface copy or dimensional labels. Illustrative decorative object, not validated functional engineering geometry.

Master: 2000x1500; ratio 4:3. Mobile: Keep critical object inside central 65%; do not crop screen/hand/contact point; use separate 4:3 crop or stack whole illustration below text.

Alt: Decorative alt empty when adjacent text supplies purpose; actual work/product images have specific localized descriptive alt.

G17 - Generic miniature boat concept

Source: generate-illustration. Containers: P01.hero, P07.related, P09.example.

Photorealistic professional TiLADYS marketing illustration, an original simple white PLA miniature toy boat with modest cabin. Three-quarter product angle, fine visible layers and ordinary scale; use original generic design rather than a copied branded benchmark model Soft studio/daylight from upper left; gentle fill, controlled shadows; no neon glow. Materials: matte white PLA, layered deck and hull. Brand palette navy #071837, restrained cyan #7be0df, white and charcoal. Clean composition, consistent colour grading, realistic scale and perspective. No logos, watermarks, baked interface copy or dimensional labels. Respect model rights; no assumption that benchmark model licences permit resale or redistribution.

Master: 2000x1500; ratio 4:3. Mobile: Keep critical object inside central 65%; do not crop screen/hand/contact point; use separate 4:3 crop or stack whole illustration below text.

Alt: Decorative alt empty when adjacent text supplies purpose; actual work/product images have specific localized descriptive alt.

G18 - Porous decorative sphere concept

Source: generate-illustration. Containers: P10.hero.

Photorealistic professional TiLADYS marketing illustration, a white PLA open-pattern decorative sphere. Small desk-scale rounded object with structurally coherent connected walls; soft shadows, no complex scientific interpretation Soft studio/daylight from upper left; gentle fill, controlled shadows; no neon glow. Materials: white layered PLA, natural openings. Brand palette navy #071837, restrained cyan #7be0df, white and charcoal. Clean composition, consistent colour grading, realistic scale and perspective. No logos, watermarks, baked interface copy or dimensional labels.

Master: 2000x1500; ratio 4:3. Mobile: Keep critical object inside central 65%; do not crop screen/hand/contact point; use separate 4:3 crop or stack whole illustration below text.

Alt: Decorative alt empty when adjacent text supplies purpose; actual work/product images have specific localized descriptive alt.

G19 - Black and white PLA spools

Source: generate-illustration. Containers: P04.materials.

Photorealistic professional TiLADYS marketing illustration, two ordinary unbranded black and white PLA spools. Side-by-side balanced product crop, spool hubs and filament winding plausible; adequate room for separate text labels Soft studio/daylight from upper left; gentle fill, controlled shadows; no neon glow. Materials: plastic spools and layered filament winding. Brand palette navy #071837, restrained cyan #7be0df, white and charcoal. Clean composition, consistent colour grading, realistic scale and perspective. No logos, watermarks, baked interface copy or dimensional labels.

Master: 2100x1400; ratio 3:2. Mobile: Keep critical object inside central 65%; do not crop screen/hand/contact point; use separate 4:3 crop or stack whole illustration below text.

Alt: Black and white filament spools.

G20 - Sketch and caliper helper photo

Source: generate-illustration. Containers: P11.3d-helper, D04.print-helper.

Photorealistic professional TiLADYS marketing illustration, an unlabelled hand sketch of a simple printable object beside a digital caliper and small neutral print. Top-down or 30-degree overhead view, paper right, object left, caliper lays safely in foreground; no precise dimensions baked in Soft studio/daylight from upper left; gentle fill, controlled shadows; no neon glow. Materials: paper pencil lines, metal caliper, PLA. Brand palette navy #071837, restrained cyan #7be0df, white and charcoal. Clean composition, consistent colour grading, realistic scale and perspective. No logos, watermarks, baked interface copy or dimensional labels. Exact dimensional diagrams/labels are native SVG overlays separately verified; imagegen must not supply measurements.

Master: 2000x1500; ratio 4:3. Mobile: Keep critical object inside central 65%; do not crop screen/hand/contact point; use separate 4:3 crop or stack whole illustration below text.

Alt: Decorative alt empty when adjacent text supplies purpose; actual work/product images have specific localized descriptive alt.

G21 - Local support workspace illustration

Source: generate-illustration. Containers: P05.local-card, P10.local-card.

Photorealistic professional TiLADYS marketing illustration, a clean modest PC workbench with monitor, chair and basic tools. Wide straightforward desk view, no office signage/address; calm daylight, visually practical Soft studio/daylight from upper left; gentle fill, controlled shadows; no neon glow. Materials: ordinary wood, black devices, neat tools. Brand palette navy #071837, restrained cyan #7be0df, white and charcoal. Clean composition, consistent colour grading, realistic scale and perspective. No logos, watermarks, baked interface copy or dimensional labels. Illustration, not a photograph of the actual premises.

Master: 2400x1500; ratio 16:10. Mobile: Keep critical object inside central 65%; do not crop screen/hand/contact point; use separate 4:3 crop or stack whole illustration below text.

Alt: Decorative alt empty when adjacent text supplies purpose; actual work/product images have specific localized descriptive alt.

G22 - Remote support illustration

Source: generate-illustration. Containers: P05.remote-card, P10.remote-card.

Photorealistic professional TiLADYS marketing illustration, a laptop with neutral monitor-link icon placeholder beside a headset. Three-quarter simple worktable view, no remote-access software logos or private data, scene light enough for thumbnail Soft studio/daylight from upper left; gentle fill, controlled shadows; no neon glow. Materials: matte laptop, fabric headset, wood/slate. Brand palette navy #071837, restrained cyan #7be0df, white and charcoal. Clean composition, consistent colour grading, realistic scale and perspective. No logos, watermarks, baked interface copy or dimensional labels.

Master: 2400x1500; ratio 16:10. Mobile: Keep critical object inside central 65%; do not crop screen/hand/contact point; use separate 4:3 crop or stack whole illustration below text.

Alt: Decorative alt empty when adjacent text supplies purpose; actual work/product images have specific localized descriptive alt.

G23 - Education decorative still life

Source: generate-illustration. Containers: P10.education.

Photorealistic professional TiLADYS marketing illustration, ordinary unbranded engineering/electronics notebooks and books near a small plant. Close desk crop, stacked plain spines, notebook and pencil; educational context without readable degree claims Soft studio/daylight from upper left; gentle fill, controlled shadows; no neon glow. Materials: paper/bookcloth, muted wood. Brand palette navy #071837, restrained cyan #7be0df, white and charcoal. Clean composition, consistent colour grading, realistic scale and perspective. No logos, watermarks, baked interface copy or dimensional labels. No fake university diploma, readable book title or certification.

Master: 2400x1500; ratio 16:10. Mobile: Keep critical object inside central 65%; do not crop screen/hand/contact point; use separate 4:3 crop or stack whole illustration below text.

Alt: Decorative alt empty when adjacent text supplies purpose; actual work/product images have specific localized descriptive alt.

G24 - Setup/software service thumbnail

Source: generate-illustration. Containers: P05.setup-card.

Photorealistic professional TiLADYS marketing illustration, a clean laptop with a neutral blank blue desktop screen. Three-quarter device shot, screen ready for actual licensed software capture inserted later, no invented OS logos Soft studio/daylight from upper left; gentle fill, controlled shadows; no neon glow. Materials: black glass/metal and neutral desk. Brand palette navy #071837, restrained cyan #7be0df, white and charcoal. Clean composition, consistent colour grading, realistic scale and perspective. No logos, watermarks, baked interface copy or dimensional labels.

Master: 2000x1250; ratio 16:10. Mobile: Keep critical object inside central 65%; do not crop screen/hand/contact point; use separate 4:3 crop or stack whole illustration below text.

Alt: Decorative alt empty when adjacent text supplies purpose; actual work/product images have specific localized descriptive alt.

G25 - Troubleshooting thumbnail

Source: generate-illustration. Containers: P05.troubleshooting-card.

Photorealistic professional TiLADYS marketing illustration, a laptop with a simple neutral diagnostic blue screen placeholder. Device close-up, no gibberish error text; illustrate problem-solving visually without

making screen content evidence Soft studio/daylight from upper left; gentle fill, controlled shadows; no neon glow. Materials: ordinary laptop glass and matte frame. Brand palette navy #071837, restrained cyan #7be0df, white and charcoal. Clean composition, consistent colour grading, realistic scale and perspective. No logos, watermarks, baked interface copy or dimensional labels.

Master: 2000x1250; ratio 16:10. Mobile: Keep critical object inside central 65%; do not crop screen/hand/contact point; use separate 4:3 crop or stack whole illustration below text.

Alt: Decorative alt empty when adjacent text supplies purpose; actual work/product images have specific localized descriptive alt.

G26 - Cleaning thumbnail

Source: generate-illustration. Containers: P05.cleaning-card, D04.pc-example.

Photorealistic professional TiLADYS marketing illustration, a careful hand using a soft brush around an unpowered open PC fan. Close three-quarter hardware shot, realistic fingers and contact position; components plausible; no powered wet cleaning Soft studio/daylight from upper left; gentle fill, controlled shadows; no neon glow. Materials: brush bristles, metal heatsink, matte plastic. Brand palette navy #071837, restrained cyan #7be0df, white and charcoal. Clean composition, consistent colour grading, realistic scale and perspective. No logos, watermarks, baked interface copy or dimensional labels. Do not depict dangerous powered repairs, inappropriate liquid or impossible hand/tool geometry.

Master: 2000x1250; ratio 16:10. Mobile: Keep critical object inside central 65%; do not crop screen/hand/contact point; use separate 4:3 crop or stack whole illustration below text.

Alt: Decorative alt empty when adjacent text supplies purpose; actual work/product images have specific localized descriptive alt.

G27 - SSD/RAM upgrade thumbnail

Source: generate-illustration. Containers: P05.upgrade-card.

Photorealistic professional TiLADYS marketing illustration, a compatible-looking RAM module and SSD near a PC motherboard. Clean macro with correctly oriented slots, no made-up vendor lettering; do not imply compatibility with every device Soft studio/daylight from upper left; gentle fill, controlled shadows; no neon glow. Materials: PCB, copper contacts, matte heat spreader. Brand palette navy #071837, restrained cyan #7be0df, white and charcoal. Clean composition, consistent colour grading, realistic scale and perspective. No logos, watermarks, baked interface copy or dimensional labels.

Master: 2000x1250; ratio 16:10. Mobile: Keep critical object inside central 65%; do not crop screen/hand/contact point; use separate 4:3 crop or stack whole illustration below text.

Alt: Decorative alt empty when adjacent text supplies purpose; actual work/product images have specific localized descriptive alt.

G28 - Backup/data thumbnail

Source: generate-illustration. Containers: P05.backup-card.

Photorealistic professional TiLADYS marketing illustration, an external drive beside a laptop with blank safe screen. Simple desk close-up, cable physically connected plausibly, room for native data-transfer icon overlay Soft studio/daylight from upper left; gentle fill, controlled shadows; no neon glow. Materials: brushed drive case, black cable, glass screen. Brand palette navy #071837, restrained cyan #7be0df, white and charcoal. Clean composition, consistent colour grading, realistic scale and perspective. No logos, watermarks, baked interface copy or dimensional labels. No personal filenames or fake encryption certification.

Master: 2000x1250; ratio 16:10. Mobile: Keep critical object inside central 65%; do not crop screen/hand/contact point; use separate 4:3 crop or stack whole illustration below text.

Alt: Decorative alt empty when adjacent text supplies purpose; actual work/product images have specific localized descriptive alt.

G29 - Remote service thumbnail

Source: generate-illustration. Containers: P05.remote-card-small.

Photorealistic professional TiLADYS marketing illustration, a laptop showing an unlabelled monitor-link visual and headset. Quiet straight-on three-quarter crop, native icon may be added separately instead of baked in Soft studio/daylight from upper left; gentle fill, controlled shadows; no neon glow. Materials: matte hardware/soft desk. Brand palette navy #071837, restrained cyan #7be0df, white and charcoal. Clean composition, consistent colour grading, realistic scale and perspective. No logos, watermarks, baked interface copy or dimensional labels.

Master: 2000x1250; ratio 16:10. Mobile: Keep critical object inside central 65%; do not crop screen/hand/contact point; use separate 4:3 crop or stack whole illustration below text.

Alt: Decorative alt empty when adjacent text supplies purpose; actual work/product images have specific localized descriptive alt.

G30 - Generic website-preview device carrier

Source: generate-illustration. Containers: P03.gallery-concepts, P01.selected-work-placeholder.

Photorealistic professional TiLADYS marketing illustration, a neutral laptop/tablet displaying a blank bright rectangle. Three-quarter screen perspective with generous inset, suitable for later authentic screenshot composite; no website company names Soft studio/daylight from upper left; gentle fill, controlled shadows; no neon glow. Materials: matteblackframe,softneutral desk. Brand palette navy #071837, restrained cyan #7be0df, white and charcoal. Clean composition, consistent colour grading, realistic scale and perspective. No logos, watermarks, baked interface copy or dimensional labels. Real gallery uses actual screenshots; this is illustration carrier only.

Master: 2400x1500; ratio 16:10. Mobile: Keep critical object inside central 65%; do not crop screen/hand/contact point; use separate 4:3 crop or stack whole illustration below text.

Alt: Decorative alt empty when adjacent text supplies purpose; actual work/product images have specific localized descriptive alt.

G31 - Template preview carrier

Source: generate-illustration. Containers: P03.templates, P08, P09, D09.

Photorealistic professional TiLADYS marketing illustration, desktop and mobile neutral device frames with blank preview inserts. Clean front-ish paired frames, desktop behind mobile right; use actual owned template screens later Soft studio/daylight from upper left; gentle fill, controlled shadows; no neon glow. Materials: black frame/clean glass, light neutral backdrop. Brand palette navy #071837, restrained cyan #7be0df, white and charcoal. Clean composition, consistent colour grading, realistic scale and perspective. No logos, watermarks, baked interface copy or dimensional labels. Do not invent Framer products or availability; no fake purchase price.

Master: 2400x1500; ratio 16:10. Mobile: Keep critical object inside central 65%; do not crop screen/hand/contact point; use separate 4:3 crop or stack whole illustration below text.

Alt: Decorative alt empty when adjacent text supplies purpose; actual work/product images have specific localized descriptive alt.

R01 - Founder portrait

Source: real-user-photo. Containers: P01.founder-teaser, P10.portrait, D13.account, D12.content-editor.

Acquire a real approved photograph of Vladyslav Titov. Small 240x300 desktop crop, natural head/shoulders, focal point stored; 480x600 or 960x1200 derivative; preserve actual identity. Never replace with the generated stranger shown in some mockups.

Master: Original photo plus 960x1200 max webmaster; ratio 4:5. Mobile: Use the same 4:5 head-and-shoulders portrait crop at compact display size; retain face, identity and focal point. Never use a generic 4:3 hero crop.

Alt: Vladyslav Titov, founder of TiLADYS.

R02 - Mongoose Signs website

Source: actual-portfolio-media. Containers: P01.selected-work, P03.gallery (website projects only), P06.grid, P07.detail, D08.project-editor.

Acquire actual approved project screenshots/photos from current published Project media. Keep screen/photo intact, produce 16:10 thumbnail and original detail gallery; no customer-private data; redact only with reviewed derivatives. Generated cafe/physio/plant-shop examples do not replace these verified project names.

Master: Source-dependent; target 1600-wide detail; ratio source plus 16:10. Mobile: Use reviewed 16:10 thumbnails with the actual page/photo focal point; detail view retains complete original or accessible zoom. Do not crop essential evidence.

Alt: Localized actual project title and subject.

R03 - DonetsFit website

Source: actual-portfolio-media. Containers: P01.selected-work, P03.gallery (website projects only), P06.grid, P07.detail, D08.project-editor.

Acquire actual approved project screenshots/photos from current published Project media. Keep screen/photo intact, produce 16:10 thumbnail and original detail gallery; no customer-private data; redact only with reviewed derivatives. Generated cafe/physio/plant-shop examples do not replace these verified project names.

Master: Source-dependent; target 1600-wide detail; ratio source plus 16:10. Mobile: Use reviewed 16:10 thumbnails with the actual page/photo focal point; detail view retains complete original or accessible zoom. Do not crop essential evidence.

Alt: Localized actual project title and subject.

R04 - DonetsFit Calculator

Source: actual-portfolio-media. Containers: P01.selected-work, P03.gallery (website projects only), P06.grid, P07.detail, D08.project-editor.

Acquire actual approved project screenshots/photos from current published Project media. Keep screen/photo intact, produce 16:10 thumbnail and original detail gallery; no customer-private data; redact only with reviewed derivatives. Generated cafe/physio/plant-shop examples do not replace these verified project names.

Master: Source-dependent; target 1600-wide detail; ratio source plus 16:10. Mobile: Use reviewed 16:10 thumbnails with the actual page/photo focal point; detail view retains complete original or accessible zoom.

Do not crop essential evidence.

Alt: Localized actual project title and subject.

R05 - Dual-boot workstation

Source: actual-portfolio-media. Containers: P01.selected-work, P03.gallery (website projects only), P06.grid, P07.detail, D08.project-editor.

Acquire actual approved project screenshots/photos from current published Project media. Keep screen/photo intact, produce 16:10 thumbnail and original detail gallery; no customer-private data; redact only with reviewed derivatives. Generated cafe/physio/plant-shop examples do not replace these verified project names.

Master: Source-dependent; target 1600-wide detail; ratio source plus 16:10. Mobile: Use reviewed 16:10 thumbnails with the actual page/photo focal point; detail view retains complete original or accessible zoom. Do not crop essential evidence.

Alt: Localized actual project title and subject.

R06 - ETIT/VPU virtual tour

Source: actual-portfolio-media. Containers: P01.selected-work, P03.gallery (website projects only), P06.grid, P07.detail, D08.project-editor.

Acquire actual approved project screenshots/photos from current published Project media. Keep screen/photo intact, produce 16:10 thumbnail and original detail gallery; no customer-private data; redact only with reviewed derivatives. Generated cafe/physio/plant-shop examples do not replace these verified project names.

Master: Source-dependent; target 1600-wide detail; ratio source plus 16:10. Mobile: Use reviewed 16:10 thumbnails with the actual page/photo focal point; detail view retains complete original or accessible zoom. Do not crop essential evidence.

Alt: Localized actual project title and subject.

R07 - TiLADYS business card

Source: actual-portfolio-media. Containers: P01.selected-work, P03.gallery (website projects only), P06.grid, P07.detail, D08.project-editor.

Acquire actual approved project screenshots/photos from current published Project media. Keep screen/photo intact, produce 16:10 thumbnail and original detail gallery; no customer-private data; redact only with reviewed derivatives. Generated cafe/physio/plant-shop examples do not replace these verified project names.

Master: Source-dependent; target 1600-wide detail; ratio source plus 16:10. Mobile: Use reviewed 16:10 thumbnails with the actual page/photo focal point; detail view retains complete original or accessible zoom. Do not crop essential evidence.

Alt: Localized actual project title and subject.

R08 - JobAdys

Source: actual-portfolio-media. Containers: P01.selected-work, P03.gallery (website projects only), P06.grid, P07.detail, D08.project-editor.

Acquire actual approved project screenshots/photos from current published Project media. Keep screen/photo intact, produce 16:10 thumbnail and original detail gallery; no customer-private data; redact only with reviewed derivatives. Generated cafe/physio/plant-shop examples do not replace these verified project names.

Master: Source-dependent; target 1600-wide detail; ratio source plus 16:10. Mobile: Use reviewed 16:10 thumbnails with the actual page/photo focal point; detail view retains complete original or accessible zoom. Do not crop essential evidence.

Alt: Localized actual project title and subject.

R09 - Real Framer templates

Source: actual-product-media. Containers: P03.template-card, P08.catalogue, P09.detail, D09.product-editor.

Acquire screenshots and preview URLs for each actual owned published Framer template. Desktop/mobile/page previews, ordered images and thumbnail; title/price/licence confirmed separately. Studio/Nova/Fork mockup names not assumed real products.

Master: 1600-2400-wide originals; ratio 16:10 plus mobile. Mobile: Keep critical object inside central 65%; do not crop screen/hand/contact point; use separate 4:3 crop or stack whole illustration below text.

Alt: Decorative alt empty when adjacent text supplies purpose; actual work/product images have specific localized descriptive alt.

R10 - Real completed print photos

Source: real-print-photo. Containers: P04.gallery, P06.grid, P07.print-detail, D08, D04.job-files.

Acquire genuine photos of the user's completed black/white prints. Same consistent desk lighting where possible; before/during/after metadata kept private; approved public derivative selected explicitly. No generated geometry sold as completed client work; check model licence and customer permission.

Master: Original plus 1600-wide; ratio 4:3. Mobile: Keep critical object inside central 65%; do not crop screen/hand/contact point; use separate 4:3 crop or stack whole illustration below text.

Alt: Decorative alt empty when adjacent text supplies purpose; actual work/product images have specific localized descriptive alt.

R11 - Real downloadable model previews

Source: actual-model-render. Containers: P04.model-card, P08, P09, D09.

Render the actual owned/licensed STL or other supported model in CAD/rendering software. Consistent camera/background; show actual shape and scale where verified; native SVG dimensions separately. A pretty generated product does not prove the downloadable mesh matches.

Master: 2000x1500; ratio 4:3. Mobile: Keep critical object inside central 65%; do not crop screen/hand/contact point; use separate 4:3 crop or stack whole illustration below text.

Alt: Decorative alt empty when adjacent text supplies purpose; actual work/product images have specific localized descriptive alt.

R12 - Customer enquiry/review attachments

Source: customer-upload. Containers: D02.private-inbox, D15.review, P11.optional-approved-review.

Original customer files preserved privately with safe derivatives. Private preview authorization; review photos public only after moderation/permission; labels/filenames treated as text. Never regenerate customer evidence or publish private inbox attachments.

Master: Bounded original; safe image variants; ratio source. Mobile: Keep critical object inside central 65%; do not crop screen/hand/contact point; use separate 4:3 crop or stack whole illustration below text.

Alt: Decorative alt empty when adjacent text supplies purpose; actual work/product images have specific localized descriptive alt.

N01 - Profile placeholder and empty states

Source: native-vector-css. Containers: D01.profile, D13.account, all-empty-states.

Native initials VT circle, simple icon and explanatory HTML text. Neutral circle tint, no mocked face; labelled icon and empty-state CTA. The generated profile faces are discarded.

Master: SVG/CSS; ratio 1:1. Mobile: Not applicable: use responsive native CSS, SVG or data layout; do not rasterize text or controls..

Alt: Decorative alt empty when adjacent text supplies purpose; actual work/product images have specific localized descriptive alt.

N02 - Charts, sketches, measurements and PDF previews

Source: native-data-visual. Containers: D11.estimator, D16.reports, P11.dimension-helper, D06.pdf-preview.

Render actual data with native UI/SVG; PDF thumbnail from actual DE-first snapshot. No AI chart numbers; precise unit labels; German page first, English second; chart legend and table fallback. Do not copy tax rates, amounts or fake Swiss addresses from mockups.

Master: Responsive SVG / actual PDF; ratio 16:9. Mobile: Not applicable: use responsive native CSS, SVG or data layout; do not rasterize text or controls..

Alt: Decorative alt empty when adjacent text supplies purpose; actual work/product images have specific localized descriptive alt.

Appendix C. Acceptance matrix

All results are planned, not executed during this analysis. Use isolated synthetic records, mocked mail and controlled upload fixtures. P0 means security/financial/data-integrity release blocker; P1 means core-flow release blocker; P2 means feature and quality verification.

ID	Priority / area	Scenario	Expected result
A01	P0 / Confidentiality	Anonymous requests a client, job, invoice, PDF or private file endpoint	401/403 or login; no private data in response, cache or search index.
A02	P0 / Files	Substitute an attachment ID from a different enquiry	Reject 404/403 after checking the parent relationship; no unrelated bytes.
A03	P0 / Publication	Request draft or archived project media, including resize URLs	Public 404; neither original nor derivative is served.
A04	P0 / CSRF	Submit a cookie-authenticated mutation from an untrusted or missing Origin	Reject under the exact origin policy; no mutation.
A05	P0 / Issuance	Two concurrent issue requests followed by a retry after response loss	One invoice number, immutable PDF and issuance event.
A06	P0 / Draft lock	Save a draft concurrently with issuance	Issued data and bytes remain unchanged; late save returns conflict.
A07	P0 / Payments	Concurrent partial payments together exceed the remaining balance	Only permissible amount commits; conflicting request fails, no hidden negative balance.
A08	P0 / Payments	Retry the same payment after response loss	Return the same payment using idempotency; no duplicate ledger entry.
A09	P0 / Snapshots	Edit client, catalogue and bank settings after invoice issuance	Issued PDF, checksum, number and totals stay unchanged.
A10	P0 / Invoice DE-first	Preview and issue a short bilingual invoice	Complete German page 1 and English page 2; same invoice identifier and amounts.
A11	P0 / Invoice pagination	Use long German text and one exceptionally long invoice line	Readable continuation pages; English begins on a new page after the complete German section.
A12	P0 / Legacy invoices	Deploy the German-first renderer with historical issued PDFs	Previously stored PDFs remain byte-identical; no automatic regeneration.
A13	P0 / Tax gate	Issue with unconfirmed tax settings or missing required translations	422 with actionable fields; draft preserved and no tax assumptions.
A14	P0 / Uploads	Upload a spoofed image signature or excessive pixel dimensions	Reject or quarantine before use; bounded CPU, memory and decoder work.
A15	P0 / Uploadfinalization	Finalize another actor's reservation or an oversized stored object	Reject; no attachment or client association created.
A16	P0 / SQL/XSS	Put SQL and HTML payloads in messages, names, filters and filenames	Render escaped text; no SQL execution or script; safe download filename.

ID	Priority / area	Scenario	Expected result
A17	P0 / Restore	Restore an isolated database and matching object backup	Recover expected synthetic records, files and hashes without production overwrite.
A18	P1 / Unicode	Submit English, Ukrainian and Russian messages with newlines and emoji	Store and display the original exactly; reply language remains independent.
A19	P1 / Photos	Upload three 5MiB photos, one 5MiB+1-byte file and a 15MiB+1-byte total	Accept exact boundaries; reject excess; metadata request stays within platform limits.
A20	P1 / Mobilephotos	Select HEIC, a corrupt image and a large phone photo on weak network	Clear conversion/rejection path, bounded processing and preserved form.
A21	P1 / Contactidempotency	Double-click Send and lose the acknowledgement response	One enquiry per idempotency key; retry returns the original receipt.
A22	P1 / Mail	Persist an enquiry but fail to send the acknowledgement email	Retain enquiry and return saved status; no duplicate-producing browser failure.
A23	P1 / Reply	SMTP accepts a reply but recording its result fails	Show accepted/uncertain state and reconcile; no blind resend.
A24	P1 / Duplicateclient	Use the same email for a household or several company contacts	Offer duplicate review; no automatic merge or globally unique email failure.
A25	P1 / Expiry	Expire a session while editing a private draft	No unauthorized mutation; recover safely, reauthenticate and recheck version.
A26	P1 / Staleedit	Edit the same draft in two browser tabs	Reject stale save with revision conflict; no silent last-writer overwrite.
A27	P1 / Numbers	Create clients, jobs and invoices concurrently at Berlin New Year	Unique numbers and correct business year using atomic sequences.
A28	P1 / Catalogue	Select FROM, PERCENTAGE or MONTHLY catalogue pricing	Explicit agreed amount and valid period; never invoice a raw display string.
A29	P1 / Quote	Accept an old or expired quote revision and retry conversion	Reject or renew after review; bind exact accepted revision and convert once.
A30	P1 / Relations	Attach an invoice to a job belonging to another client	Reject the invalid relationship; no financial cross-link.
A31	P1 / SEO	Rename a published project slug several times	All old aliases redirect directly to the canonical route; no draft leak.
A32	P1 / Languages	Use a missing translation or a legacy ua language alias	Canonical Ukrainian remains uk; no false hreflang or indexable fallback duplication.
A33	P1 / Footer	Inspect the footer on desktop, tablet, phone and at 200% zoom	Retain live footer identity, verified links and readable layout without overlap.
A34	P1 / Menu	Open, scroll and close the phone menu using keyboard and navigation	Correct focus, visible content below header, no scroll bleed or hidden controls.
A35	P1 / Stateerrors	Compare an empty list with a database outage	Distinct honest states; failure does not masquerade as lost client records.

ID	Priority / area	Scenario	Expected result
A36	P1 / Models	Upload invalid STL, enormous mesh, archive or G-code	Reject or privately quarantine per separate model policy; never execute on a printer.
A37	P1 / Backuppolicy	Let provider restore history expire without an independent backup	Do not claim protection; require real backup and isolated restore evidence.
A38	P1 / Runtime	Build both applications with deployed Node 24 and the actual lockfile	Required existing/new checks and builds pass with compatible settings.
A39	P1 / Publication	Create a portfolio draft from a completed job containing private notes	Exclude confidential data; publish only explicitly approved media derivatives.
A40	P1 / Bounds	Exercise rate limits across replicas and behind a shared public IP	Shared counters work with bounded limits and understandable retry guidance.
A41	P2 / Externalstore	Open an unavailable template preview or purchase listing	Honest unavailable state; no invented stock, sales or synchronization.
A42	P2 / TasksDST	Schedule an appointment in an ambiguous or nonexistent local DST time	Explicit timezone resolution; list and calendar agree.
A43	P2 / Review	Moderate a review with private photos and malicious text	Check publication permission; safely render original; never publish private email.
A44	P2 / Reports	Report invoices issued in a different period from their payments, with mixed currencies	Keep period bases and currency totals separate; no invented profit.
A45	P2 / Exports	Export CSV containing formula payloads and a large file archive	Neutralize formulas; use bounded private expiring downloads and safe archive paths.
A46	P2 / Estimate	Compare 20% markup with 20% margin and convert grams to kilograms	EUR15 cost gives EUR18 with markup or EUR18.75 with margin; clearly labelled.
A47	P2 / Accessibility	Use keyboard, screen reader, zoom and reduced-motion preferences	Operable controls, visible focus, announced states and no reflow overlap.
A48	P2 / Performance	Load realistic public media on a phone	Measure LCP, INP and CLS; use responsive derivatives with reserved aspect ratios.
A49	P1 / Identity	Observe Neon managed-auth tables while custom login is active	Do not migrate identity or enable public admin registration without a separate decision.
A50	P0 / Secrets	Inspect browser bundles, exports and diagnostic logs	No database URL, auth secret, SMTP password, bypass token or unintended private data.
A51	P0 / Cancel/correction	Request invoice cancellation when payments already exist	Preserve original artifact and payment history; reviewed correction and balance treatment.
A52	P1 / IssuedPDFmissing	Download an issued invoice whose original PDF bytes are missing	Specific conflict/operator recovery state; never rebuild from mutable current data.